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Integrated analysis of dynamic changes in volatile and non-volatile compounds in Chuan Chenpi (*Citrus reticulata* 'Dahongpao') during aging by GC-IMS and UPLC-MS/MS

Jialiang Zou [a,b], Huangsheng Zhang [a,b], Liang Zhao [a,b], Huanhuan Xu [a,b], Lin Chen [a,b], Hongping Chen [a,b], Fu Wang [a,b], Yuan Hu [a,b,*], Youping Liu [a,b,*]

[a] School of Pharmacy, Chengdu University of Traditional Chinese Medicine, Chengdu 611137, China
[b] Lab for Innovation & Effective Uses of Chinese Drug Germplasm Resources, Chengdu University of Traditional Chinese Medicine, Chengdu 611137, China

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ABSTRACT

Chuan Chenpi (CCP), the peel of the ancient Chinese citrus variety *Citrus reticulata* 'Dahongpao', enhances its edible and medicinal value through aging. This study focuses on CCP samples aged 1–8 years, aiming to explore the dynamic changes during aging. Although aging induced noticeable changes in color, Fourier Transform Infrared (FTIR) analysis revealed no significant alterations in functional groups. Volatile and non-volatile metabolomic analyses indicated that aging primarily affects metabolite content rather than types, and both composition and antioxidant activity reached relatively high levels in the fourth year. The experimental results illustrate that the improvement in CCP quality induced by aging does not follow a linear pattern but exhibits fluctuations. The findings verify that the fourth year serves as a critical turning point for CCP's quality. This study clarifies the dynamic variation of chemical components in CCP during aging, thereby providing a scientific basis for the selection of its aging duration.

1. Introduction

Chenpi (Citri Reticulatae Pericarpium, CRP) refers to the dried mature peel of *C. reticulata* Blanco and its cultivated varieties. It has been utilized as an herbal medicine for thousands of years, demonstrating excellent efficacy in treating digestive and respiratory disorders (Cai et al., 2025). Owing to its rich content of various bioactive compounds such as flavonoids, alkaloids, and essential oils, CRP also possesses considerable edible value, serving as a valuable source of health foods and dietary supplements (Liu et al., 2021). Additionally, due to its distinctive flavor, it is employed as a condiment. The 2020 edition of the *Pharmacopoeia of the People's Republic of China* (Chinese Pharmacopoeia, *ChP*) lists four main cultivated varieties under the Chenpi entry: *C. reticulata* 'Chachi' (Guang Chenpi, GCP), *C. reticulata* 'Dahongpao' (Chuan Chenpi, CCP), *C. reticulata* 'Unshiu', and *C. reticulata* 'Tangerina' (Chinese Pharmacopoeia Commission, 2020). *C. reticulata* 'Dahongpao' was once the principal cultivated citrus variety in Sichuan and Chongqing. With a cultivation history of more than 4000 years in China, it is renowned as a global gene bank for red mandarins and has played a significant role in the history of citrus cultivation in China. The dried mature peel of *C. reticulata* 'Dahongpao' is the well-known Sichuan authentic medicinal material CCP, which was historically famous nationwide for its large production volume and superior quality (Luo et al., 2023; Xu et al., 2001). CCP is characterized by its affordable price and notable health benefits, with significant effects in disease prevention and treatment, making it highly favored by local practitioners. In recent decades, however, the cultivation area of *C. reticulata* 'Dahongpao' has sharply declined due to competition from emerging citrus varieties, leading to a substantial reduction in CCP supply and an insufficient availability of high-quality CRP (Li et al., 2023; Li, Zhao, et al., 2024). Furthermore, the lack of systematic research on the aging process of CCP has resulted in unclear trends in the transformation of its active compounds during aging, making the process disorganized. Therefore, conducting systematic studies on the aging of CCP is of

* Corresponding authors at: School of Pharmacy, Chengdu University of Traditional Chinese Medicine, Chengdu 611137, China.
E-mail addresses: huyuan@cdutcm.edu.cn (Y. Hu), youpingliu@cdutcm.edu.cn (Y. Liu).

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particular importance for ensuring the sustainable and stable supply of this valuable herbal medicine, while also expanding options for high-quality health foods and dietary supplements.

Aging is a distinctive feature of CRP, as it is believed to enhance both its flavor and health-promoting functions. Over time, the active components in CRP undergo transformation and release through physicochemical processes and microbial activity, thereby altering its flavor and medicinal or edible value (Liu et al., 2025; Shorbagi et al., 2022). In contrast, GCP has been more extensively researched in terms of aging. Studies have shown that polymethoxyflavones increase (Liang et al., 2022; Zeng et al., 2023), while synephrine content decreases during aging (Wei et al., 2013). Regarding volatile organic compounds (VOCs), aging leads to a reduction in D-limonene in essential oils, a decrease in low-molecular-weight components, and an increase in high-molecular-weight constituents (Wang et al., 2013). Hu et al. (2025) further demonstrated that novel compounds such as thymol are generated in later stages of aging, contributing to the improvement of CCP's flavor profile. Research by Wang et al. (2023) indicated that microbial activity on the surface of CRP, dominated by species such as *Aspergillus niger*, promotes the accumulation of flavonoids through metabolic transformation during the aging process. While GCP is widely regarded as the most renowned and highest-quality CRP, making it an ideal source for health foods and dietary supplements, previous studies have revealed that CCP actually contains higher levels of key active compounds such as hesperidin, nobiletin, and tangeretin compared to GCP (Song et al., 2017; Yang, 2022; Zhang, 2018), highlighting its significant potential and promising prospects. Moreover, GCP is primarily used as a seasoning and raw material for tea production. Well-aged GCP holds greater collectible value, commands high prices, and is limited by geographical origin (Deng et al., 2024; Li, Zhao, et al., 2024). Under these circumstances, GCP can no longer meet the public's demand for daily health foods and medicinal applications, and its high cost is not affordable for the general population. This further underscores the significant market potential for high-quality CCP in health foods and medicinal uses. Therefore, in-depth research on CCP and its application to fill this market gap is particularly urgent. CRP is widely applied in the food industry, with its primary use as a raw material for tea beverages. Aging can enhance the flavor and aroma of tea (Lan et al., 2025). Additionally, various products such as Jiuzhi Chenpi (Fu et al., 2022), Chenpi jelly (Peng et al., 2022), and Chenpi candies have been developed. In cooking, CRP can also be used as a seasoning for congee and braised meats (Yu et al., 2018), and signature dishes such as Chenpi chicken and Chenpi beef have been created (Zhang et al., 2020). This demonstrates the extensive applications of CRP in the food sector, highlighting the immense application potential of CCP in this field.

Determining the aging duration of CRP has long been a challenging task. Traditionally, this relied on the professional expertise of practitioners (Li et al., 2024), a method that is highly subjective and not always accurate. With advancements in scientific technology and the development of analytical instruments with diverse functionalities, the aging period and its differentiation of CRP can now be assessed objectively. During the aging process, the color, odor, and chemical composition of CCP undergo varying degrees of change. These alterations can be precisely differentiated using various modern instruments. The electronic eye (*E*-eye), a type of visual analyzer, simulates the human eye's visual perception of samples to identify different specimens. Characterized by its simplicity, speed, and no need for pre-treatment, it has proven effective in identifying counterfeit aging years and determining the geographical origin of CRP (Qin et al., 2024; Zhang et al., 2023). Similarly, the electronic nose (*E*-nose) is designed to mimic the olfactory systems of humans and animals, capable of recognizing, analyzing, and assessing odors. Its application across various regions (Chen et al., 2025; Liu et al., 2025), different phenological stages (Zhang et al., 2025), and during processing stages (Lan et al., 2025) demonstrates recognition of this methodology. While the E-nose can determine general odor profiles, it cannot perform detailed qualitative or quantitative analysis of specific compounds. Methods based on chromatography and mass spectrometry, such as Gaschromatography-mass spectrometry (GC–MS), GC-IMS, and UPLC-MS/MS, represent classical approaches for differentiation (Li, Zhao, et al., 2024). These techniques can detect subtle changes in chemical compounds and have even led to the development of targeted analytical methods focused on specific classes of compounds. Furthermore, other spectroscopic analysis methods, such as FTIR, have also played a significant role in comparing different samples (Li et al., 2024). By integrating multiple analytical instruments, it is possible to accurately identify changes during the aging process of CCP and achieve precise differentiation of its aging duration.

In this study, a combination of analytical techniques including *E*-eye, E-nose, GC-IMS, and UPLC-MS/MS was employed to comprehensively and systematically characterize the dynamic changes in color, odor, and chemical composition during the aging process of CCP aged 1–8 years (CCPY1–8). The research elucidated the transformation patterns during CCP aging, identified the optimal aging duration, and provided a scientific basis for standardizing the aging procedure. These findings hold significant implications for the comprehensive utilization and development of *C.reticulata* 'Dahongpao' peel and its application in health food products.

2. Materials and methods

2.1. Materials and reagents

All samples of Chuan Chenpi (CCP) used in our research, with aging periods ranging from one to eight years (designated as CCPY1 to CCPY8), were provided by Sichuan Wanwu Bencao Health Technology Co., Ltd. The raw materials were collected from Chaoyang Town, Shizhong District, Neijiang City. Citrus samples were collected from the same cultivar, *C. reticulata* 'Dahongpao', grown in a single planting base. Fruits were uniformly harvested annually from late December to early January of the following year during 2017–2024. After washing, the fruits were manually peeled, and the peels were naturally sun-dried to obtain fresh CCP. Subsequently, the fresh CCP was packed into cloth bags, placed in wooden boxes, and aged on the second floor of a cool, dry, and well-ventilated warehouse (to prevent moisture damage). The aging temperature and humidity varied with seasons (annual average temperature: 15–24 °C; annual average humidity: 72%–80%) without any artificial intervention. The CCP samples were turned over and sun-dried annually in August. All samples were subjected to three biological replicates.

Methanol, acetonitrile, and formic acid were purchased from Merck Drugs & Biotechnology, USA, and were chromatographically pure. Methanol, potassium bromide (KBr) and potassium persulfate ($K_2S_2O_8$) were purchased from Chengdu Kelon Chemicals Co. 2-Butanone, 2-Pentanone, 2-Hexanone, 2-Heptanone, 2-Octanone, and 2-Nonanone were all supplied by Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). 1,1-diphenyl-2-trinitrophenylhydrazine (DPPH) and 2,2′-azino-bis-3-ethylbenzothiazoline-6-sulfonic acid (ABTS) were purchased from Shanghai Macklin Biochemical Technology Co (Shanghai, China). Total antioxidant capacity (T-AOC) assay kit (ferric ion reducing antioxidant capacity, FRAP) was purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, Jiangsu Province, China). All reagents were of analytical grade, with the exception of KBr which was of spectral grade.

2.2. Analysis of the physicochemical properties of CCP from different aging years

2.2.1. Total aerobic plate counting and Escherichia coli counting

The total aerobic plate count and *Escherichia coli* count were tested by Weifang Hairun Huachen Testing Technology Co., Ltd. (Shandong, China), which holds China Metrology Accreditation (CMA),

International Laboratory Accreditation Cooperation - Mutual Recognition Arrangement (ILAC-MRA), and China National Accreditation Service for Conformity Assessment (CNAS) qualifications. The determination of total aerobic plate count was carried out in accordance with the national standard GB 4789.2–2022. Precisely weigh 25 g of sample powder and add it to 225 mL of sterile normal saline, homogenize at 8000 r/min for 2 min to prepare a 1:10 dilution. After serial gradient dilution (from 10^{-1} to 10^{-6}), select 3 appropriate dilutions, aspirate 0.1 mL of each and evenly spread it on nutrient agar medium plates (3 parallel samples for each dilution). Incubate at a constant temperature of 36 °C ± 1 °C for 48 h ± 2 h before counting, and express the results as colony-forming units per gram (CFU/g). The determination of *Escherichia coli* was performed with reference to the industry standard SN/T 0169–2010. Take the aforementioned 1:10 dilution and inoculate it into lactose bile salt fermentation tubes using the Most Probable Number (MPN) method, incubate at 36 °C ± 1 °C for 24 h ± 2 h for primary fermentation. Pick the gas-producing bacterial liquid and inoculate it on Eosin Methylene Blue agar medium for isolation and culture. Select typical colonies for Gram staining and biochemical identification, then refer to the MPN index table based on the number of positive tubes, and express the results as MPN/g.

2.2.2. Determination of flavonoid content

The quantification of hesperidin, nobiletin, and tangeretin was performed according to the method specified in the *ChP* (2020 Edition, Volume I) (Chinese Pharmacopoeia Commission, 2020) for the determination of CRP. Specifically, approximately 0.2 g of the powdered sample was accurately weighed into a stoppered conical flask. Then, 25 mL of methanol was precisely added, the flask was sealed and weighed. The mixture was subjected to ultrasonication (300 W, 40 kHz) for 45 min. After cooling, the flask was weighed again, and the weight loss was compensated by adding methanol. The solution was mixed well, filtered through a 0.22 μm microporous membrane, and stored in an injection vial for HPLC analysis. The HPLC analysis was conducted using an Agilent 1260 system (Agilent Technologies Inc., California, USA) equipped with an Agilent SB-C_{18} column (250 mm × 4.6 mm, 5 μm). The mobile phase consisted of acetonitrile (A) and water (B). The elution program was set as follows: initially 78% B, maintained for 10 min, then linearly changed to 52% B over 10 min, and held for another 10 min. The flow rate was 1.0 mL/min, the column temperature was maintained at 30 °C, and the injection volume was 5 μL. Detection wavelengths were set at 283 nm for hesperidin and 330 nm for nobiletin and tangeretin, using a UV detector. In addition, the CCP extract was also utilized for the determination of total flavonoid content. Quantification was achieved using a standard curve ($y = 0.0072\times + 0.0684$, $R^2 = 0.9994$) generated with different concentrations of hesperidin.

2.2.3. FTIR analysis

The sample powder was mixed with an appropriate amount of dry KBr and ground in an agate mortar until a homogeneous mixture was obtained. This mixture was then pressed into a transparent pellet under a pressure of 15 kN for 10 s. The pellet was analyzed using an FTIR-650 Fourier Transform Infrared Spectrometer (Tianjin Gangdong Sci. & Tech. Co., Ltd., Tianjin, China). Measurements were carried out in the mid-infrared range of 4000 to 400 cm^{-1}, with each spectrum representing an average of 32 scans.

2.2.4. Color variation of CCP with different aging years based on E-eye analysis

An appropriate amount of powdered sample was placed in a sample cup and leveled by compression. Color measurement was then performed using a non-contact imaging spectrophotometer (MetaVue VS3200, Shanghai X-Rite Color Technology Co., Ltd., Shanghai, China). The L*, a*, and b* values in the CIE L*a*b* color space were recorded. In this system, L* represents lightness, with higher values indicating greater brightness; a* represents the green-red axis, with higher values indicating more red tone; and b* represents the blue-yellow axis, with higher values indicating more yellow tone.

2.2.5. Odor variation of CCP with different aging years based on E-nose analysis

An appropriate amount of powdered sample was placed in a 20 mL headspace vial. Two needles were inserted through the vial septum – one connected to the sensor array and the other to an activated carbon filter - for analysis using an AIT-N1200 electronic nose (Beijing iSensortalk Technology Co., Ltd., Beijing, China). Each measurement lasted for 1 min, with data from 55 to 58 s being recorded for odor analysis. After each sample analysis, the system was purged with filtered clean air for 2 min.

2.3. GC–IMS analysis of CCP from different aging years

The analysis was performed based on a previously reported method with minor modifications(Wang, Wang, et al., 2023), using a GC-IMS instrument (FlavourSpec®, Gesellschaft für Analytische Sensorsysteme mbH, Germany) equipped with a MXT-WAX column (15 m × 0.53 mm, 1 μm film thickness). Specifically, 0.5 g of the sample was placed in a 20 mL headspace vial and incubated at 60 °C for 15 min with constant agitation at 500 rpm. Thereafter, 200 μL of the headspace gas was automatically injected into the instrument for analysis. The column oven and drift tube temperatures were set at 60 °C and 45 °C, respectively, while the injector temperature was maintained at 80 °C. High-purity nitrogen (≥99.999%) was used as both carrier and drift gas. The drift gas flow was set at 150 mL/min. The carrier gas flow was programmed as follows: initially held at 2 mL/min for 2 min, then increased linearly to 10 mL/min from 2 to 10 min, further raised to 100 mL/min from 10 to 20 min, and finally maintained for an additional 30 min, resulting in a total run time of 50 min.

A standard mixture of six ketones (2-butanone, 2-pentanone, 2-hexanone, 2-heptanone, 2-octanone, and 2-nonanone) was analyzed to establish calibration curves for retention time and retention index. The retention index of each target compound in the samples was then calculated based on its retention time. Qualitative analysis was performed by searching and matching these indices against the built-in NIST 2020 GC retention index database and the IMS migration time database within the VOCal data processing software (version 0.4.10). Furthermore, plugins such as Reporter and Gallery Plot in the VOCal software were utilized to generate three-dimensional (3D) spectra, two-dimensional (2D) topographic plots, differential spectra, and fingerprint plots of the VOCs. These visualizations were employed for comparative analysis of VOCs across different samples.

2.4. Non-targeted metabolomics analysis of CCP from different aging years

The non-targeted metabolomics analysis was performed following a previously established method with minor modifications (Zou et al., 2024). Briefly, the samples were vacuum freeze-dried using a freeze dryer (Ningbo Xinzhi Bio-technology Co., Ltd., Ningbo, Zhejiang Province, China), ground to a fine powder, and 30 mg of the powder was precisely weighed. Then, 1500 μL of a − 20 °C pre-cooled 70% methanol-water solution containing an internal standard was added for extraction. After centrifugation, the supernatant was collected and filtered through a microporous membrane (0.22 μm pore size). The resulting filtrate was stored in injection vials for UPLC-MS/MS analysis.

Chromatographic separation was achieved using a Waters ACQUITY UPLC HSS T3 column (1.8 μm, 2.1 mm × 100 mm). The mobile phase consisted of solvent A (ultrapure water containing 0.1% formic acid) and solvent B (acetonitrile containing 0.1% formic acid). The gradient elution program was set as follows: initial 5% B, increased to 65% B over 5.0 min, then raised to 99% B in 1.0 min and held for 1.5 min, followed by a reduction to 5% B within 0.1 min and a re-equilibration period of

2.4 min. The total run time was 10.0 min. The flow rate was 0.4 mL/min, the column oven temperature was maintained at 40 °C, and the injection volume was 4.0 μL.

The analysis was performed on an AB TripleTOF 6600 mass spectrometer. The electrospray ionization (ESI) source parameters were set as follows: source temperature, 550 °C, ion spray voltage, 5000 V (ESI+) / -4000 V (ESI-); nebulizer gas (GS1), heater gas (GS2), and curtain gas were set at 50, 60, and 35 psi, respectively. The collision energy (CE) was set at 10 V (ESI+) / -10 V (ESI-) for the MS1 survey scan and 30 V (ESI+) / -30 V (ESI-) for the MS2 product ion scan. The mass range was acquired from *m/z* 50 to 1250 for MS1 and from m/z 25 to 1250 for MS2.

The raw mass spectrometry data were first converted to mzML format using ProteoWizard. Peak detection, alignment, and retention time correction were then performed with the XCMS package. The processed peaks were annotated by searching against a combination of an in-house laboratory database, integrated public databases, in-silico prediction libraries, and via the metDNA method. Finally, we retained metabolites with a composite identification score above 0.5 and a coefficient of variation (CV) in quality control (QC) samples below 0.5. Ions detected in both positive and negative ionization modes were merged, retaining the one with the highest identification confidence and the lowest CV, to yield the final dataset.

2.5. Analysis of antioxidant activity of CCP from different aging years

Approximately 0.2 g of the powdered sample was ultrasonically extracted (SB25-12D, Ningbo Xinyi Ultrasonic Equipment Co., Ltd., Ningbo, Zhejiang Province, China) with 10 mL of 70% methanol at 26 ± 2 °C for 1 h (100 W, 50 Hz). The extract was then centrifuged at 4000 rpm (TDZS-WS, Hunan Xiangyi Laboratory Instrument Development Co., Ltd.) for 20 min, filtered, and diluted to various concentrations. The resulting solutions were used within one week for antioxidant activity studies.

Using the method of Zou et al. (2024) with slight modifications, we evaluated the ABTS and DPPH radical scavenging activities of extracts at various concentrations (0.1, 0.5, 1.0, 2.0, 3.0, 4.0 mg/mL). All results were replicated three times and expressed as semi-inhibitory concentration (IC_{50}). Total antioxidant activity (FRAP) was determined according to the manufacturer's kit protocol, with results expressed as milligrams of ferrous sulfate per gram of dry weight sample (mg $FeSO_4$ g^{-1} DW).

2.6. Data analysis

All bar charts, line graphs, and analysis of variance (ANOVA) presented in this article were generated using GraphPad Prism 10.23 (GraphPad Software, USA), with a *P*-value <0.05 considered statistically significant. Principal component analysis (PCA), cluster heatmaps, Venn diagrams, K-means analysis, pie charts, and flavor wheels were created using Metware Cloud, a free online platform for data analysis (https://cloud.metware.cn). Box plots and raincloud plots were produced via the Personalbio Genescloud platform (https://www.genescloud.cn). Radar charts were plotted using Origin 2022 professional graphing software. Metabolites were annotated using the KEGG database (Compound: https://www.kegg.jp/kegg/compound/; Pathway: http://www.kegg.jp/kegg/pathway.html).

3. Results and discussion

3.1. Changes in physicochemical properties of CCP during aging

To evaluate the safety of CCP for consumption after long-term storage, the total aerobic plate count and *Escherichia coli* levels of CCPY1–CCPY8 were determined (Table S1). The results indicated that the total aerobic plate count remained relatively high within the first five years of aging, while from the sixth year onward, it stabilized at a low level. Despite the relatively higher counts observed in the early aging stage, they were still far below the general limit for food products (typically $\leq 10^5$ CFU/g). Regarding *Escherichia coli* detection, this indicator was undetectable in all CCP samples (<3.0 MPN/g). Collectively, these findings demonstrate that CCP maintains high edible safety throughout the entire aging period. The variation in microbial counts during aging may be attributed to the annual sun-drying process conducted in August. However, the specific microbial taxa undergoing succession and the underlying succession patterns during CCP aging require further investigation using microbiome sequencing technologies. This aspect will be the focus of our subsequent research endeavors.

Flavonoids represent a significant class of bioactive compounds in CRP. In this study, we quantified the changes in total flavonoids and three specific flavonoid monomers during the aging process of CCP. During aging, the total flavonoid content increased continuously over the first four years, peaking at 9.73% in the fourth year, after which it declined but remained stable and ultimately higher than the initial level (Fig. 1A). The content of nobiletin consistently increased, starting at 0.64% in the first year and reaching a maximum of 0.72% by the sixth year, before declining slightly, though it remained elevated compared to earlier stages (Fig. 1B). Similarly, tangeretin content increased with aging, peaking at 0.29% in the fourth year, then decreasing to the initial level by the eighth year (Fig. 1C). Unexpectedly, the hesperidin content exhibited continuous fluctuations throughout the aging process (Fig. D), which may be associated with metabolic activities of microorganisms on the peel surface. This observation is supported by Wang, Hu, et al. (2023), who confirmed that microbial activity influences flavonoids levels in aging CRP. Mamy et al. (2022) and Fu et al. (2025) demonstrated that solid-state and liquid fermentation of CRP using *Aspergillus niger* CGMCC 3.6189 and *Bacillus* spp., respectively, significantly enhanced flavonoid content, including increases in hesperidin, nobiletin, and tangeretin, thereby accelerating flavonoids accumulation. Further studies by Mamy et al. (2025) demonstrated that solid-state fermentation increased the flavonoid content in citrus peel powder by elevating the levels of phenylalanine, tyrosine, and methionine, as well as the activities of tyrosine ammonia-lyase and β-D-glucosidase. Phenylalanine ammonia-lyase (PAL) is the initial enzyme in polyphenol biosynthesis, which catalyzes the conversion of phenylalanine to cinnamic acid; the latter is then converted to p-coumaric acid under the action of cinnamate 4-hydroxylase (C4H). Meanwhile, tyrosine ammonia-lyase (TAL) can directly convert tyrosine to p-coumaric acid. Subsequently, 4-coumarate-CoA ligase (4CL) catalyzes the conversion of p-coumaric acid to p-coumaroyl-CoA, which is further transformed into naringenin chalcone, naringenin and other flavonoids (Milke et al., 2018). In addition, *Aspergillus* species can decompose other flavonoid components to promote the synthesis of hesperidin and hesperetin (Dai et al., 2024). Gao et al. (2014) found that enzymes produced by microorganisms can perform structural modifications on flavonoid components, including hydrolyzing the glycosyl moieties of flavonoid glycosides to form aglycones, glycosylating flavonoid aglycones to improve the water solubility of flavonoids, and hydroxylating flavonoids to enhance their biological activity.

FTIR identifies the molecular structure and chemical functional groups of a sample by measuring its absorption of infrared light at different wavelengths. Here, we performed FTIR analysis on CCP aged for 1–8 years (Fig. 1E). Characteristic absorption peaks were observed at 3403 cm^{-1} and 2924 cm^{-1}, corresponding to O—H and C—H stretching vibrations, respectively. The absorption near 1737 cm^{-1} is attributed to C═O stretching vibrations, which typically indicate the presence of aldehydes, ketones, or carboxylic acids. The signal at 1631 cm^{-1} is assigned to the asymmetric stretching vibration of free carboxylate groups (Liu et al., 2022). The peak at 1057 cm^{-1} is associated with in-plane C—H bending and C—H stretching vibrations within the ring structures of pyranose and furanose (Singh et al., 2025). Notably, aging appears to have no significant effect on the functional groups or molecular structure of CCP, suggesting that the aging process does not alter

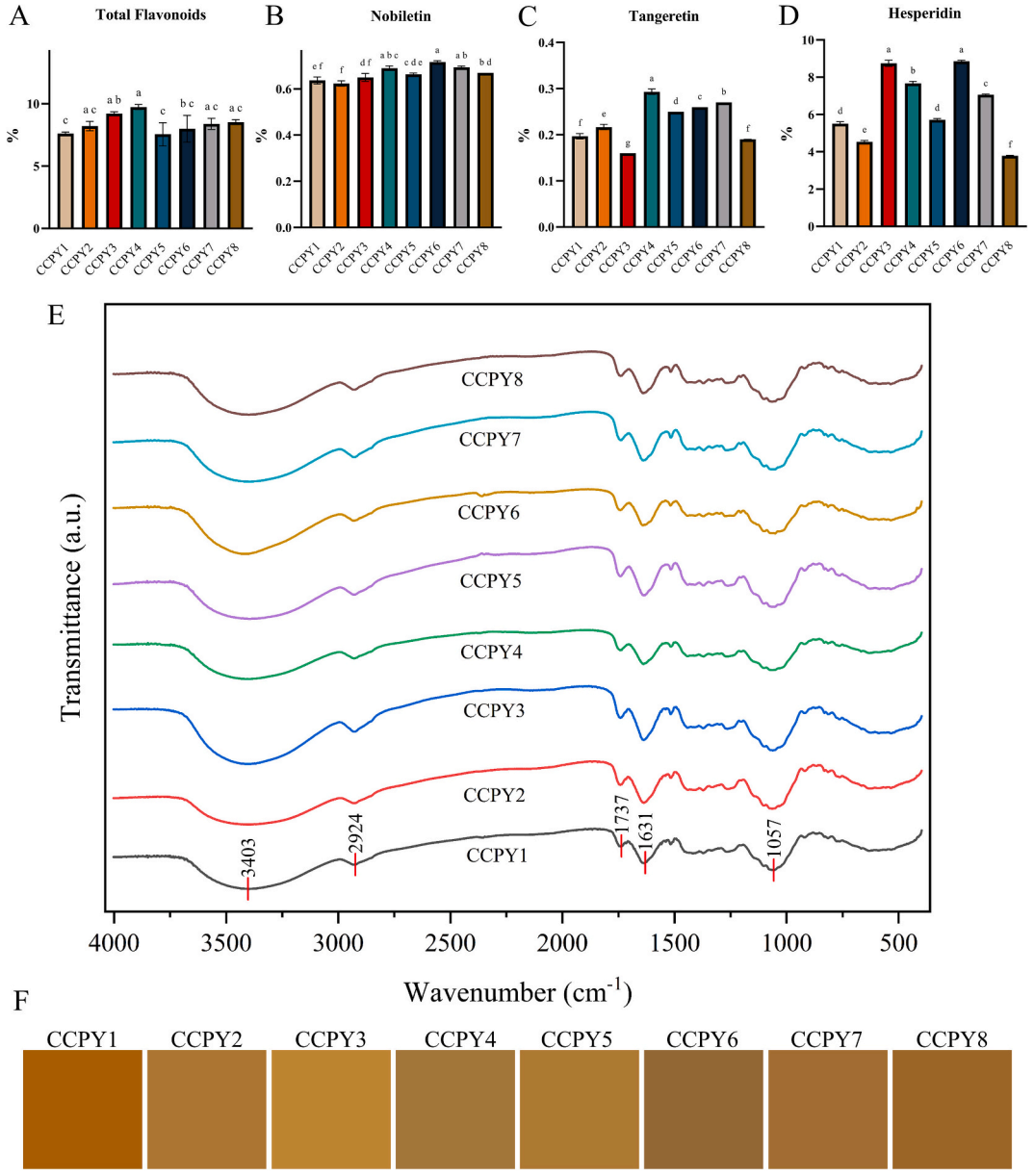


Fig. 1. Changes in flavonoid content, FTIR, and color of CCP during aging. (A) Total flavonoids. (B) Nobiletin. (C) Tangeretin. (D) Hesperidin. (E) FTIR analysis. (F) Sample color analyzed by E-eye. Different letters in the graphs indicate significant differences ($P < 0.05$). All tests were repeated three times ($n = 3$).

the types of compounds present or generate new functional group-containing compounds.

The surface color of CRP darkens with prolonged aging, making it a key indicator for distinguishing CRP of different aging years. We observed changes in the surface color of CCP during the aging process and therefore employed an *E*-eye system to quantitatively assess these color variations. As shown in Fig. 1F and Table S2 the surface color remained relatively bright in the initial years of aging but gradually deepened over time. The brightness (L* value) of the peel surface exhibited a darkening trend, while the red (a* value) and yellow (b* value) components gradually diminished. In fact, the aging of CRP is a solid-state fermentation process, and the color changes result from both enzymatic and non-enzymatic browning reactions (Hsu et al., 2024). These changes are not limited to the peel surface but are also reflected in the color of the corresponding tea infusion, with longer aging years leading to a darker infusion color (Liu et al., 2024). PCA (Fig. S1) further classified the samples into three distinct groups: CCPY1 formed one separate group; CCPY2, CCPY3, and CCPY4 clustered into a second group, showing similar and relatively stable colors with minor variations; while samples from the later five aging years (CCPY5–CCPY8) aggregated into a third group, indicating that the surface color had stabilized by this stage.

The *E*-nose, designed to mimic human olfactory perception, was employed to broadly assess odor variations in CCP across different aging years. The PCA (Fig. S2A) of E-nose shows that all CCPs exhibit similar odor characteristics, which are difficult to distinguish. Aging did not change the base flavor of CCPs, and further flavor analysis requires the use of other instrument methods. A line chart illustrating the response intensity of different sensors (Fig. S2B) revealed that the W5S sensor (high sensitivity to nitrogen oxides) produced the strongest response across all samples, suggesting a potentially higher content of nitrogen oxides, followed by the W1S sensor (sensitive to methyl groups). Notably, these findings diverge from those reported by Liu et al. (2025). While variations in the botanical origin of the CRP may contribute to this discrepancy, differences in sensor sensitivity and nomenclature across E-nose manufacturers could also be influencing factors. In contrast, our

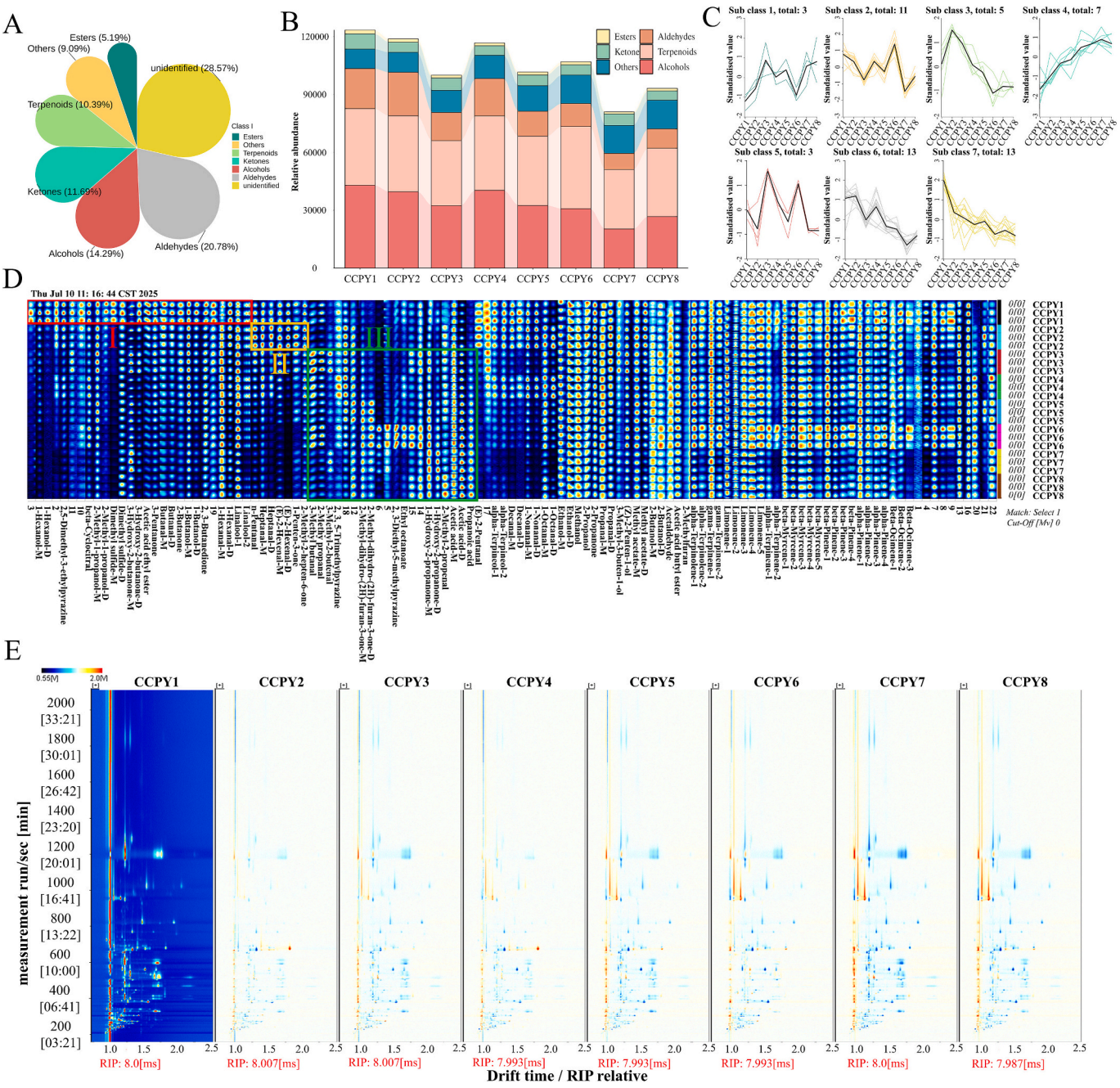


Fig. 2. Changes in VOCs of CCP during aging. (A) Pie chart of VOCs classification. (B) Stacked bar plot of VOCs content. (C) K-means analysis of VOCs. (D) VOCs fingerprint. (E) GC-IMS differential spectrum. In the fingerprint map, each row displays all selected signal peaks from one sample, while each column represents the signal peaks of the same VOCs across different samples. All tests were repeated three times (n = 3).

results align with the E-nose discrimination outcomes reported by Chen et al. (2020) and Lan et al. (2025), despite their studies focusing on CRP tea and *C. medica* L. var. sarcodactylis, respectively. Furthermore, we observed a general decline in sensor response intensity with prolonged aging. A more pronounced reduction occurred during the first four years, after which the decreasing trend moderated. This pattern indicates that aging does not alter the base odor of CCP, but sensory changes may occur as VOCs decrease.

3.2. Dynamic changes in VOCs of CCP during aging

3.2.1. Analysis of VOCs

We next employed GC-IMS to perform a comparative analysis of VOCs changes in CCP across different aging years. A total of 77 substances were detected in the eight samples (after merging duplicates); however, due to database limitations, only 55 VOCs were identified (Table S3). This highlights the necessity to expand VOCs databases to enable a more comprehensive and accurate understanding of compositional profiles in studied materials (Jia et al., 2025). As shown in Fig. 2A, the identified compounds were categorized as aldehydes (16, 20.78%), alcohols (11, 14.29%), ketones (9, 11.69%), terpenes (8, 10.39%), esters (4, 5.19%), others (7, 9.09%), and unidentified compounds (22, 28.57%). A stacked bar plot comparing the total content of all components (Fig. 2B) revealed that the VOCs content initially decreased, then increased, followed by a subsequent decline, with an overall downward trend. This result is inconsistent with the findings of Liu et al. (2025), who reported an initial increase in VOCs followed by a decrease. The discrepancy may be attributed to differences in detection

techniques.

During the aging process, some low-boiling-point VOCs gradually volatilize and dissipate, which may lead to a decrease in the total amount of VOCs ultimately detected. In general, GC-IMS exhibits higher sensitivity to polar and low-boiling compounds, but has limited detection ability for nonpolar and high-boiling substances; conversely, GC–MS shows the opposite detection characteristics. Although GC–MS is a commonly used approach for VOCs identification, its detection limit for VOCs is typically at the ppm level, whereas that of GC-IMS can reach the ppb level (Yin et al., 2021), confirming that the latter possesses a lower detection limit. Moreover, GC–MS usually requires complex sample pretreatment procedures, while GC-IMS can utilize headspace sampling, eliminating the need for tedious preprocessing. In addition, GC-IMS can efficiently and intuitively visualize differences among samples through color contour imaging (Gerhardt et al., 2019). However, the application of GC-IMS in the detection of flavor compounds in food emerged later than that of GC–MS, and its compound identification is more restricted by the coverage of available databases (Yao et al., 2024). Notably, Yao et al. (2024) found that GC–MS could identify certain alkanes, whereas GC-IMS failed to detect such compounds, mainly because alkanes have low proton affinity and are therefore difficult to ionize effectively in IMS. In a similar vein, Meng et al. (2026) reported that GC–MS shows higher sensitivity to aldehydes, while GC-IMS is more adept at detecting esters, alcohols, and ketones. It should also be noted that when using GC-IMS to detect high-boiling-point VOCs, these substances may adsorb onto the column or transfer lines due to poor volatilization. Conversely, in the analysis of low-boiling small molecules, inadequate retention time may result in insufficient resolution, leading to peak overlap or co-elution of different compounds. Furthermore, the complex matrix of food samples may also cause peak overlap or signal suppression. Nevertheless, such issues can be mitigated by optimizing sample pretreatment procedures and employing hyphenated analytical techniques (Hernández-Mesa et al., 2017). Despite these limitations, GC-IMS remains one of the most widely used techniques for analyzing VOCs. This further highlights the importance of combining multiple analytical methods to achieve a comprehensive and accurate characterization of chemical compositions and their dynamic variations (Li, Zhao et al., 2024).

Examining each compound category in detail (Fig. S3), the contents of aldehydes and alcohols also displayed an initial decrease, followed by an increase and a subsequent decline, consistent with the overall trend of total components. Notably, as aldehydes and alcohols constitute the two largest categories, their quantitative changes predominantly drive the overall compositional trend. Ester content exhibited a continuous decreasing trend, while ketones decreased substantially by the second aging year and thereafter remained relatively stable. In contrast, terpene levels fluctuated throughout the entire aging period. However, these category-level trends do not fully reveal the dynamic patterns of individual compounds. K-means clustering was therefore applied to group compounds with similar change trajectories, visualized using line charts (Fig. 2C). The analysis classified all VOCs into approximately seven clusters, among which three (Class 3, Class 4, and Class 7) showed particularly pronounced increasing or decreasing trends. More detailed box plots (Fig. S4) illustrate the actual content variation of each compound. While the general trends align with the K-means results, these plots highlight variability in absolute content levels. For instance, the relative contents of acetic acid and propanoic acid in Class 4 gradually increased with prolonged aging, whereas compounds in Class 7, such as 1-butanol, 1-hexanol, and 3-pentanone, showed a gradual decline. It is also worth noting that although linalool content decreased, it remained at relatively high levels throughout aging. Limonene, on the other hand, was present at high and stable concentrations, with no significant impact from the aging process. This observation also contrasts with previously reported findings (Liu et al., 2025), which may be attributed to multiple factors, including cultivar differences (Ma et al., 2022; Wang et al., 2024; Zhang et al., 2017) and influences of aging location and climate (He et al., 2025; Yang et al., 2022). Additionally, gamma-terpinene, alpha-terpineol, beta-myrcene, linalool, and limonene were identified as the principal VOCs in CCP, despite an observed decline in their content over time, indicating that the volatile profile of CCP undergoes continuous dynamic changes during aging. Jiang et al. (2023) analyzed the VOCs of CRP aged for six different years (2, 5, 10, 15, 20, and 25 years) and found that compounds with citrus and fruity aromas gradually decreased or even disappeared with the extension of storage time. Another study determined the contents of 17 VOCs in CRP aged for 1–3 years and revealed that their levels all decreased with aging (Shi et al., 2024). Traditional experience holds that Chenpi tea may induce body fluid loss, which may be attributed to the ‘dryness’ property of CRP, characterized by symptoms such as dry mouth, dry skin, and dry cough with scanty sputum. *Leigong Paozhi Lun*, a classic work on herbal processing in ancient China, pointed out that aging can reduce the ‘dryness’ property of CRP, and the underlying mechanism is that aging decreases the VOCs contents in CRP (Shi et al., 2024; Lan et al., 2025).

3.2.2. VOCs fingerprint analysis

As shown in the three-dimensional spectrum and top view of GC-IMS (Fig. S5), differences in VOCs among the samples can be visually observed. Subsequently, a comparative analysis of the VOCs fingerprint was performed (Fig. 2D), which displays the complete VOCs profile of each sample and highlights the variations between samples. The compounds within red frame I were most abundant in CCPY1, including 1-Hexanol, 2,5-Dimethyl-3-ethylpyrazine, beta-Cyclocitral, 2-Methyl-1-propanol, Dimethyl sulfide, 3-Hydroxy-2-butanone, Methyl acetate, 3-Pentanone, Acetaldehyde, 2-Butanone, 1-Butanol, 2,3-Butanedione, 1-Hexanal, and Linalool, among others. Those in yellow frame II peaked in CCPY2, comprising n-Pentanal, Heptanal, (E)-2-Hexenal, 1-Penten-3-one, and 2-Methyl-2-hepten-6-one, among others. The substances in green frame III were highest in specific samples from CCPY3 to CCPY8: 3-Methyl butanal, 2-Methyl propanal, and 3-Methyl-2-butenal were most abundant in CCPY3; compounds such as 2,3,5-Trimethylpyrazine peaked in CCPY4; 2-Methyl-dihydro-(2H)-furan-3-one and similar compounds were highest in CCPY5; 2,3-Diethyl-5-methylpyrazine and Ethyl octanoate, among others, peaked in CCPY6; 1-Hydroxy-2-propanone was most abundant in CCPY7; and compounds including 2-Methyl-2-propenal, Acetic acid, and Propanoic acid reached their highest levels in CCPY8. Using CCPY1 as the reference, difference maps were generated by subtracting the reference from the spectra of other samples (Fig. 2E). While the increase in red signals indicated a rise in certain volatiles like acetic acid and propanoic acid during aging, their overall contribution was more than offset by the decrease in components corresponding to the increasing blue signals.

3.2.3. Flavor analysis

We subsequently conducted Orthogonal Partial Least Squares Discriminant Analysis (OPLS-DA) by comparing CCPY2 to CCPY8 individually against CCPY1. As shown in Fig. S6 and S7, all samples were clearly distinguished. In each comparative group, 22, 40, 29, 34, 38, 40, and 38 differential VOCs were identified, respectively (VIP $>$ 1 and $P <$ 0.05). Fig. 3A and Fig. S8 display the top 20 VIP-ranked differential VOCs in each comparison, most of which were significantly upregulated in CCPY1. A flavor wheel further illustrated that these differential VOCs contribute to ‘fruity’, ‘citrus’, ‘fresh’, ‘green’, and ‘sweet’ flavor notes (Fig. 3B). Fig. 3C shows that 17 differential VOCs were shared across all comparison groups. A clustering heatmap (Fig. 3D) revealed that the majority of these shared compounds decreased in content with prolonged aging, and they were primarily associated with ‘fruity’ and ‘green’ flavors (Fig. 3E). The flavor profile of CCP differs somewhat from the known ‘citrusy’,‘spicy’,‘herbal’, and ‘pungent’ characteristics reported for GCP (Liu et al., 2024), presenting as relatively milder and exhibiting distinct flavor properties.

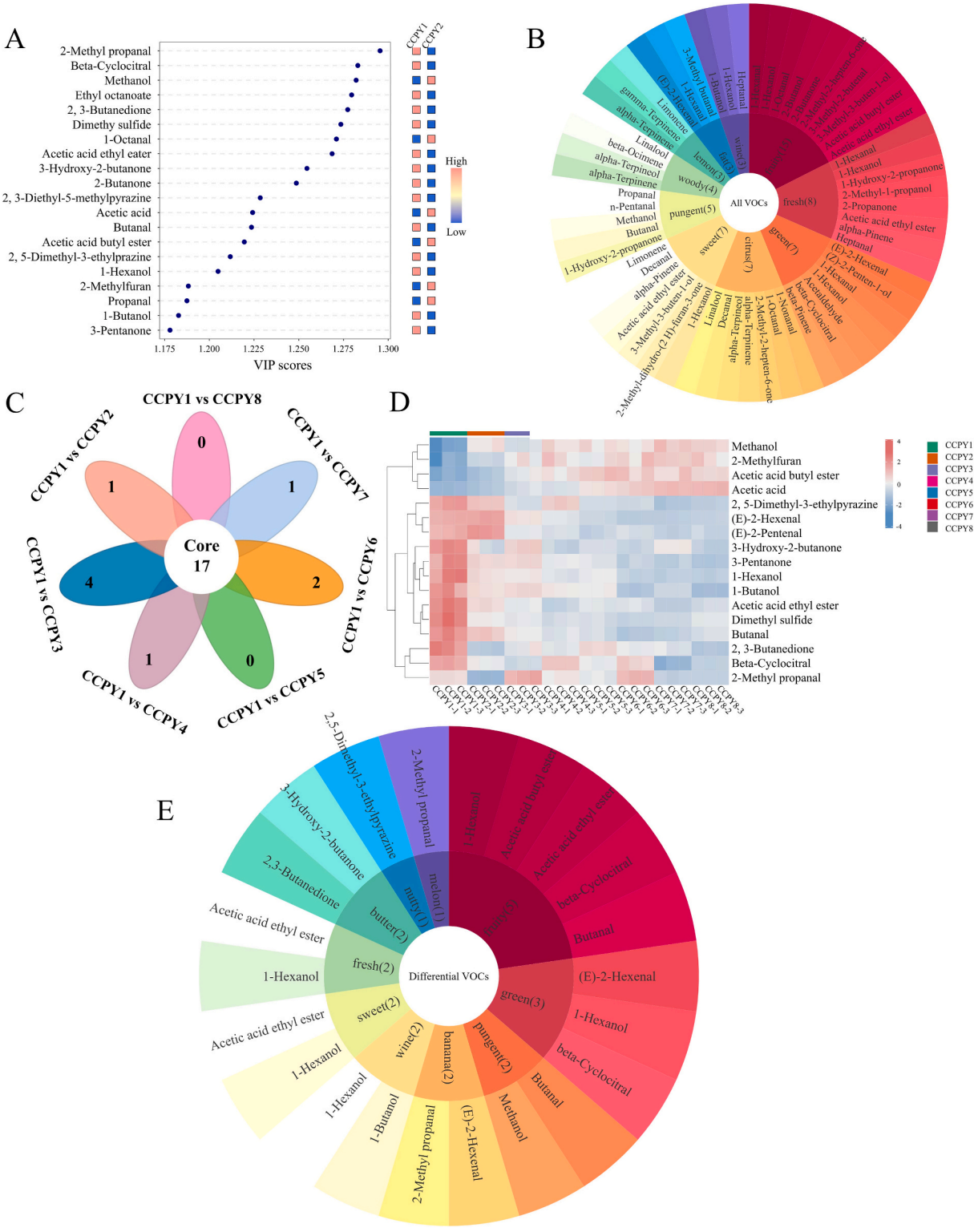


Fig. 3. Differential and flavor analysis of VOCs. (A) VIP plots and corresponding heatmaps for pairwise comparisons between CCPY1 and CCPY2. (B) Flavor wheel of all differential VOCs. (C) Venn diagram of differential VOCs from all pairwise comparisons. (D) Clustering heatmap of shared differential VOCs. (E) Flavor wheel of shared differential VOCs. Fig. A display the top 20 VOCs ranked by VIP score. The horizontal axis in VIP plots represents the VIP score, while the adjacent heatmaps illustrate the average intensity of each compound across compared groups. All tests were repeated three times (n = 3).

3.3. Dynamic changes in non-volatile components of CCP during aging

3.3.1. Metabolic profiling

Non-targeted metabolomics based on UPLC-MS/MS serves as a powerful tool for comprehensively investigating the overall metabolic profile of plants. In this study, a total of 1343 secondary metabolites were detected across eight CCP samples of different aging years. These metabolites were categorized into over 15 classes, primarily including benzene and substituted derivatives (365, 27.2%), flavonoids (173, 12.9%), and alkaloids (143, 10.6%) (Fig. 4A, Table S4). Among these,

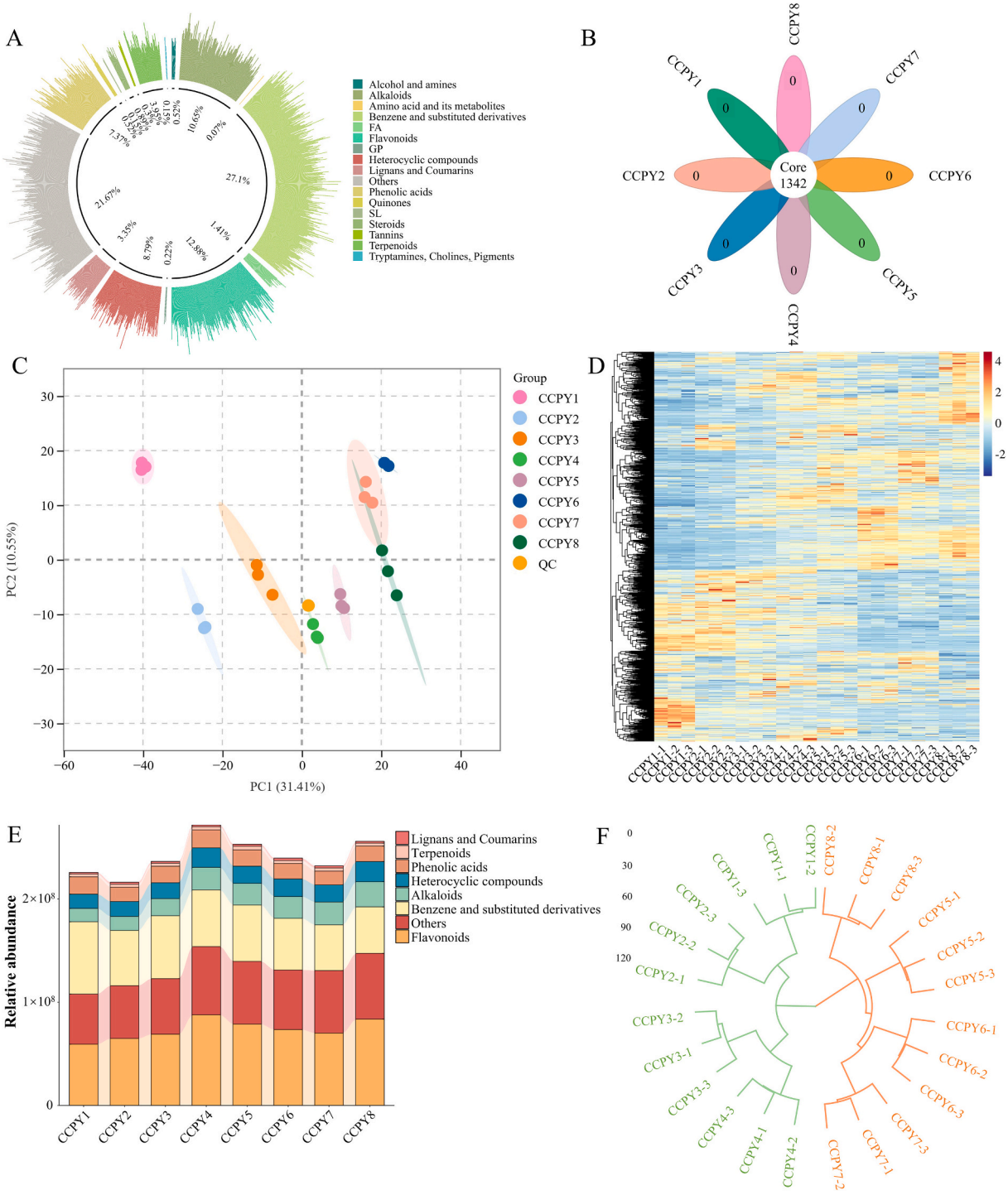


Fig. 4. Analysis of non-volatile components. (A) Pie chart of metabolite classification. (B) Venn diagram showing metabolite distribution across samples. (C) PCA score plot of metabolites. (D) Clustering heatmap of metabolites. (E) Stacked bar plot of metabolite abundance. (F) Cluster dendrogram based on metabolite relative abundance. In the Pie chart (A), the height of each peak represents the relative abundance of the corresponding metabolite. All tests were repeated three times (n = 3).

1342 metabolites were detected across all samples (Fig. 4B), indicating that the profile of metabolite types remained largely consistent throughout the aging process, which aligns with the findings from the FTIR analysis. However, distinct separation among all samples was observed in both PCA score plots and clustering heatmaps (Fig. 4C, D), confirming that aging induces substantial metabolic changes in CCP. Given the minimal change in metabolite types, these differences are likely attributable to variations in metabolite abundance. Accordingly, we compared abundance patterns across samples. A stacked bar plot (Fig. 4E) revealed a general increase in metabolite abundance during the first four years of aging, followed by a decline starting in the fifth year, and a renewed increase by the eighth year. At the class level, heterocyclic compounds and alkaloids showed continuous increases throughout aging. The content of flavonoids accounted for 26.28% in the first year of aging, showing an upward trend during the first four years and reaching 32.32% in the fourth year. Subsequently, it slightly

decreased from the fifth to the seventh year but remained between 30.20% and 31.10%. By the eighth year of aging, it increased again and peaked at 32.52%. Overall, the flavonoid content remained relatively stable from the fourth year onward, exceeding the levels observed in the initial years of aging. We subsequently constructed a cluster dendrogram based on the relative abundance of all metabolites for CCP samples aged 1 to 8 years. As shown in Fig. 4F, all samples were broadly segregated into two clusters. Cluster I comprised CCPY1 to CCPY4, indicating that the chemical profiles of the first four years of aging were more similar to each other. Within this cluster, CCPY1 and CCPY2 exhibited the highest similarity, forming a distinct sub-group, while CCPY3 and CCPY4 formed another. The remaining samples constituted Cluster II. Although these class-level trends are informative, they may not reflect the behavior of individual metabolites within each category. Therefore, we performed K-means clustering (Fig. S9), which grouped the 1343 metabolites into six distinct clusters based on their dynamic patterns. Cluster 4 contained 280 metabolites that consistently increased in abundance over time. Notably, this cluster included 42 alkaloids (29.37% of total) and 27 heterocyclic compounds (22.88% of total). In contrast, Cluster 5 comprised 254 metabolites that decreased continuously with aging, including 25 phenolic acids (25.25% of total). Cluster 6 included 188 metabolites that increased in the first four years, declined thereafter, and increased again in the eighth year, a trend closely mirroring that of flavonoids.

3.3.2. *Differential analysis*

To further investigate the effect of prolonged aging on non-volatile components in CCP, we performed pairwise OPLS-DA comparisons between CCPY2–8 and CCPY1. All comparative models showed clear separation (Fig. S10, S11). Differentially accumulated metabolites (DAMs). were then identified using the criteria of VIP $>$ 1 and FC $\geq$ 2 or FC $\leq$ 0.5. A total of 776 DAMs were detected across all comparisons, with their classifications shown in Fig. S12. As illustrated in the bar chart (Fig. 5A), the number of DAMs in each pairwise comparison ranged from 211 to 479 (132–302 up-regulated and 74–186 down-regulated). Their dynamic changes were visualized using scatter plots

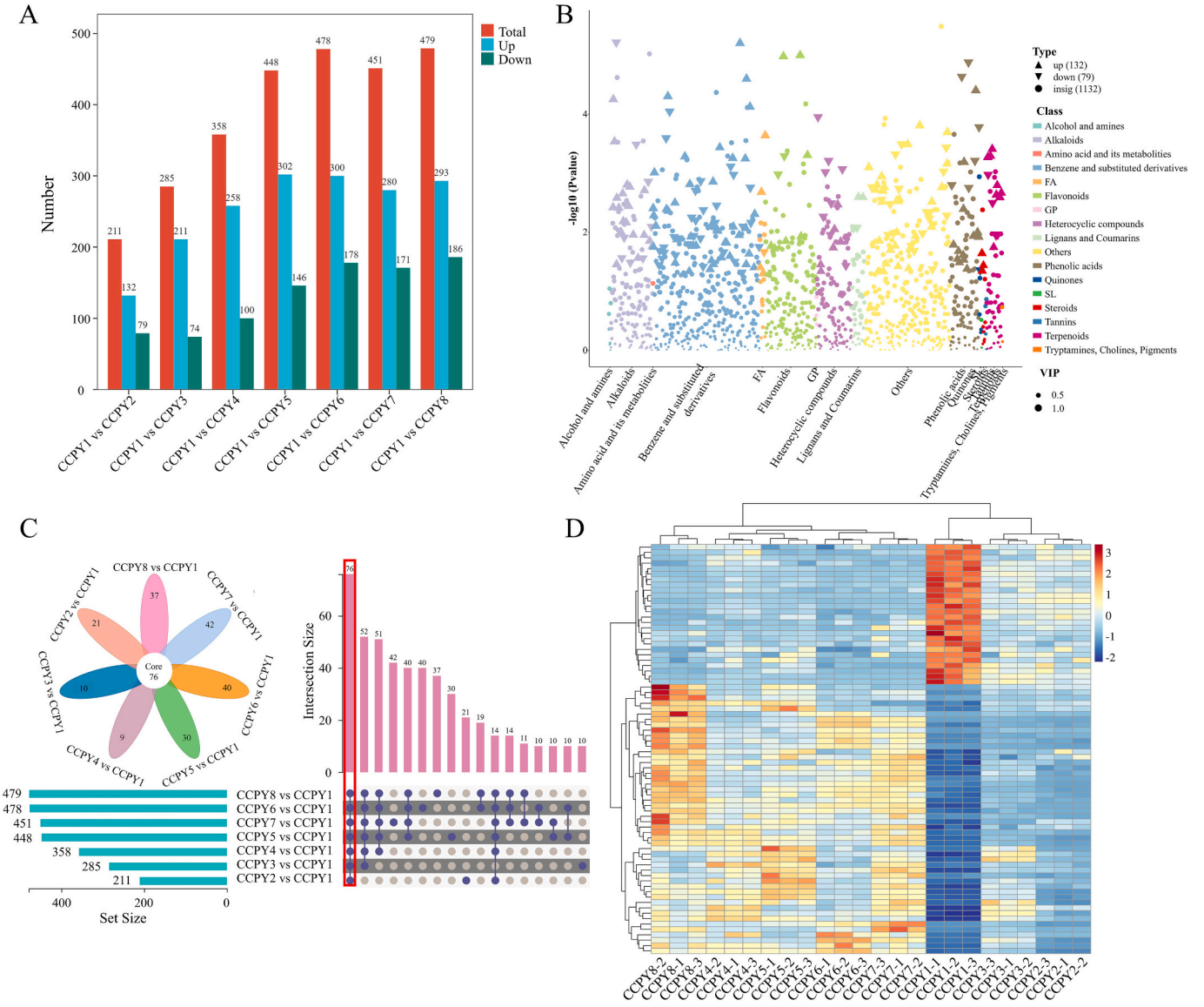


Fig. 5. Differential analysis of non-volatile components. (A) Numbers of up- and down-regulated metabolites in each comparison group. (B) Scatter plots of DAMs between CCPY1 and CCPY2. (C) Venn diagram and UpSet plot showing the distribution of DAMs. (D) Clustering heatmap of shared DAMs. In the scatter plots, the horizontal axis represents metabolite categories, with each color indicating a distinct class. Upward-pointing triangles denote up-regulated metabolites, downward-pointing triangles indicate down-regulated metabolites, and circles represent metabolites with no significant difference. The size of the symbols corresponds to the VIP value. All tests were repeated three times (n = 3).

by metabolite class (Fig. 5B, Fig. S13). The number of DAMs generally increased during the first five years of aging, rising from 211 between CCPY1 and CCPY2 to 448 between CCPY1 and CCPY5. From year 5 to year 8, the metabolic changes stabilized, with 448, 478, 451, and 479 DAMs identified in comparisons of CCPY1 with CCPY5–8, respectively, indicating a plateau in metabolic divergence. Venn diagram (Fig. 5C) analysis revealed that comparisons of CCPY1 with CCPY2–4 contained fewer unique DAMs, while later comparisons showed relatively more unique metabolites. Moreover, 76 shared DAMs were identified across all comparison groups (Fig. 5C, Table S5), which may serve as key biomarkers for distinguishing the aging years of CCP (Jia et al., 2025; Zhao et al., 2025). A clustering heatmap of the shared DAMs (Fig. 5D)

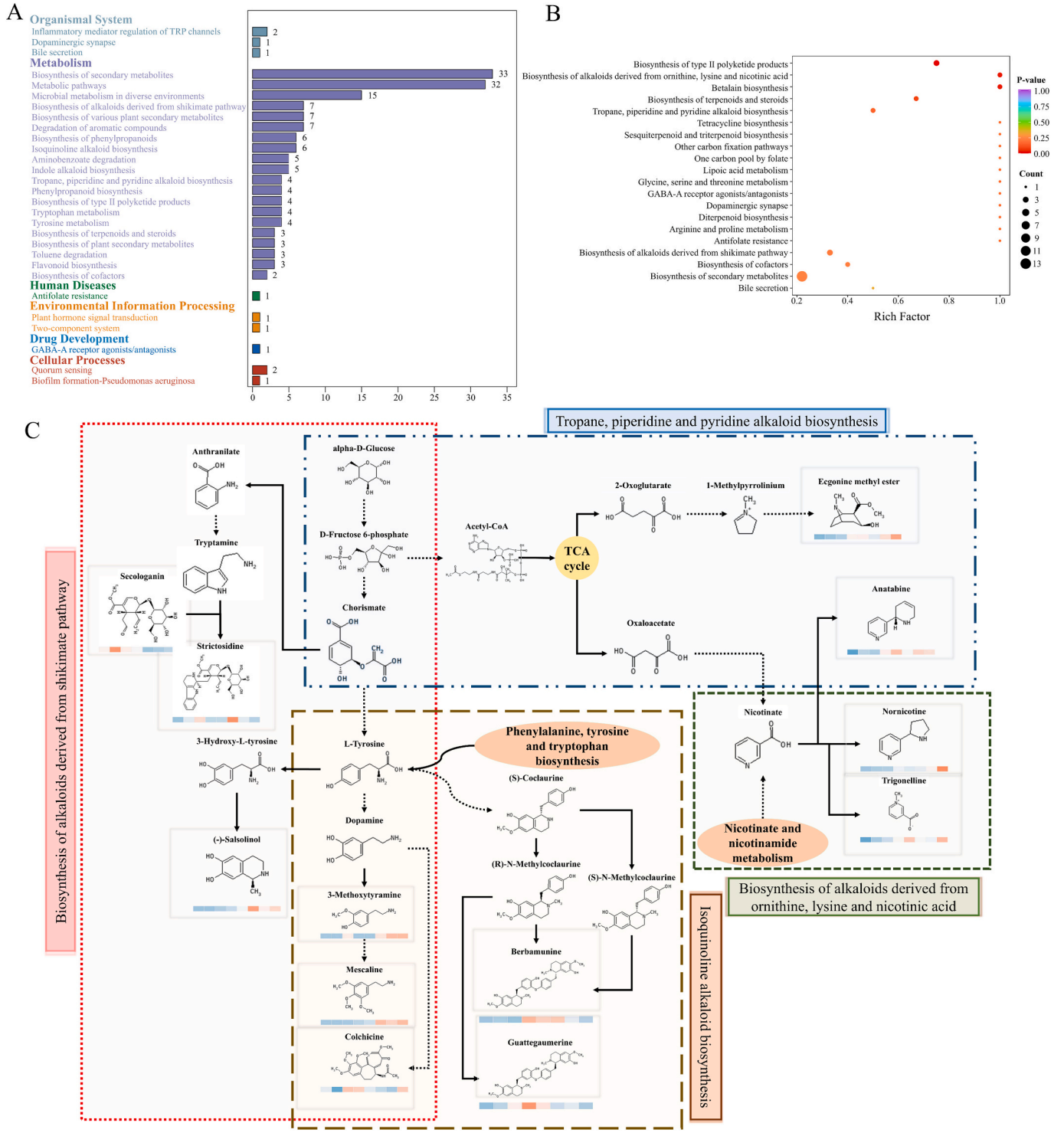


Fig. 6. KEGG functional annotation of DAMs and Alkaloid metabolic pathway map. (A) Selected metabolic pathways of DAMs. (B) KEGG enrichment bubble plots for DAMs in pairwise comparisons between CCPY1 and CCPY2. (C) Metabolic pathway map of selected Alkaloids. In the metabolic pathway map (C), solid arrows represent single-step reactions, dashed arrows indicate multi-step processes, and the heatmap colors reflect relative metabolite abundance (red for higher, blue for lower concentrations). All tests were repeated three times (n = 3). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

revealed that all samples could be broadly categorized into two major groups. CCPY2 and CCPY3 clustered together with minor differences and were subsequently grouped with CCPY1. The other cluster comprised CCPY4 and CCPY5 grouped together, CCPY6 and CCPY7 forming a distinct subgroup, and CCPY8 ultimately aggregating into the same broader category. Integrating these findings with the overall metabolite content and the trends of flavonoids, the main bioactive components in CRP, it appears that although significant changes occur during the first four years of aging, these changes generally reflect a positive qualitative transformation. In contrast, subsequent changes follow a different trajectory. In fact, changes in the composition of CRP can be induced by multiple factors. For instance, Hu et al. (2025) reported that during the process from fresh to dried and subsequent aging, CRP exhibits a deepened color, an increased flavonoid content, and a decreased volatile oil content. Meanwhile, steaming leads to the disappearance of components responsible for the fresh citrus aroma in CRP, resulting in a more intense and complex aroma; it also causes the degradation of flavonoid glycosides and an increase in bitter-tasting components (Lan et al., 2025). Additionally, delayed harvesting reduces the contents of flavonoids and phenolic acids in CRP, and high-temperature drying also induces significant transformations of VOCs and polymethoxylated flavones (Zheng et al., 2020). In summary, we found that the fourth year is a crucial turning point for CCP aging, and within a limited time frame, longer aging does not necessarily yield better quality. The study by Yang et al. (2021) also found that GCP aging is not “the longer, the better”, with the optimal quality achieved in the second year.

3.3.3. Functional annotation of DAMs

KEGG enrichment analysis helps map DAMs to relevant biological processes, providing valuable insights into the compositional changes occurring during the aging of CCP. Among the DAMs identified across comparison groups, 65 were associated with metabolic pathways, collectively annotating to 70 distinct pathways. Partial pathway information is displayed in Fig. 6A, with the complete results available in Fig. S14. Notably, 33 DAMs were annotated to the Biosynthesis of secondary metabolites, while 32 were mapped to Metabolic pathways. Interestingly, 15 metabolites were associated with Microbial metabolism in diverse environments, suggesting that environmental microorganisms or those residing on the peel surface may play an important role during aging. To date, microorganisms with the capacity to biotransform the bioactive components of CRP have been isolated and identified from its surface, including 16 bacterial genera, 22 fungal genera and 1 archaeal genus. The biotransformation of bioactive components and improvement of flavor mediated by these microorganisms represent a crucial contributor to the quality enhancement of CRP, and the microbial transformation of chemical constituents in CRP exhibits interspecific variability (Dai et al., 2024; Zhang, 2025). Dominant microbial taxa such as *Aspergillus*, *Penicillium* and *Xeromyces* play an indispensable role in the biotransformation of flavonoids and essential oils, and exert a significant impact on the dynamic changes in bioactive components (Hu et al., 2022; Wang et al., 2015; Wang, Hu, et al., 2023; Yang, 2022). Notably, the microbial biotransformation of flavonoids in CRP is primarily categorized into two types. On the one hand, glycosidases, cellulases, chitinases, xylanases and other enzymes produced by fungi during their growth and metabolism can facilitate the degradation of plant cell walls, thereby promoting the release of bound bioactive components (Sun et al., 2014; Wu et al., 2014). On the other hand, α-rhamnosidases, β-glucosidases and other enzymes derived from fungal metabolism can hydrolyze flavonoid glycosides into monoglycosides and aglycones, which further enhances the biological activity of flavonoids in CRP (Li et al., 2019; Gopalsamy Rajiv Gandhi et al., 2020). Regrettably, this study did not investigate the microbial succession during CCP aging nor dissect the functional roles of microorganisms in the aging process, which will be the key focus of our in-depth follow-up research.

We next visualized the dynamic changes in metabolic pathways during aging using bubble plots (Fig. 6B, Fig. S15). The analysis revealed that Arginine and proline metabolism, Biosynthesis of alkaloids derived from ornithine, lysine and nicotinic acid, and Tropane, piperidine and pyridine alkaloid biosynthesis were present across all comparison groups. However, enrichment in Arginine and proline metabolism was not statistically significant and involved only a limited number of compounds. In contrast, the Biosynthesis of alkaloids derived from ornithine, lysine and nicotinic acid included more metabolites and showed significant differences in two comparison groups. Among these pathways, Tropane, piperidine and pyridine alkaloid biosynthesis exhibited significant differences in six out of the seven comparison groups, representing the most markedly altered metabolic pathway during the aging process. It is also worth noting that the Biosynthesis of alkaloids derived from the shikimate pathway was identified in five comparison groups, two of which showed statistically significant differences. Our analysis revealed an interesting observation: many of the differential metabolic pathways are associated with Alkaloid biosynthesis. Alkaloids represent an important class of compounds in CRP, exhibiting a wide range of pharmacological activities such as vasoconstriction, tracheal dilation, metabolism improvement, and increased energy expenditure (Liu et al., 2021). Based on the identified compounds, we reconstructed a partial alkaloid metabolic pathway map (Fig. 6C). Within the biosynthesis pathway of Alkaloids derived from the shikimate pathway, the overall alkaloid content showed an increasing trend with prolonged aging. These compounds are synthesized via the shikimate pathway. For example, mescaline, a hallucinogenic substance derived from the cactus *Lophophora williamsii*, has shown unique efficacy in treating mental health disorders and represents an important component of psychedelic-assisted therapy (Bender, 2022; Courdavault et al., 2024; Li & Facchini, 2024). In contrast, certain alkaloids such as berbamunine and guattegaumerine, which are synthesized via the isoquinoline alkaloid biosynthesis pathway, exhibited an initial increase followed by a decrease in content. Many compounds in this class possess significant medicinal value, including morphine for analgesia, berberine for histological staining, and noscapine as an antitussive agent, highlighting their broad applicability and notable pharmacological activities. Notably, the heterologous biosynthesis of berbamunine and guattegaumerine has been successfully achieved in yeast (Payne et al., 2021).

3.4. Changes in antioxidant activity of CCP during aging

The ABTS and DPPH radical scavenging rates of CCP samples aged 1–8 years increased with rising concentration (Fig. S16). However, antioxidant activity among different samples exhibited a fluctuating pattern. Specifically, antioxidant activity declined slightly in the initial aging stage, though not significantly, and subsequently increased to levels exceeding the initial values (Fig. 7A-C). While previous studies have generally reported a continuous enhancement of antioxidant activity with prolonged aging, our results did not show a monotonic increase. Instead, we observed alternating periods of enhancement and reduction. We therefore propose that the antioxidant activity of CRP increases within a limited aging window, while exhibiting an overall increasing or fluctuating trend over a broader time scale. According to Bian et al. (2022), the antioxidant activity of GCP increased over a long-term aging period of 1–11 years, with the most pronounced increase occurring during the first five years, followed by a more gradual rise. This finding aligns to some extent with our observations, though we were unable to validate it further due to the lack of CCP samples with longer aging times. Moreover, existing literature lacks comprehensive studies on continuous temporal changes in the bioactivity of aged CRP. It is also worth noting that the variation in antioxidant activity was consistent with the trend of flavonoids. This suggests that the health-promoting effects of CRP, and indeed of many plant-derived foods and medicines, result from the combined action of multiple components.

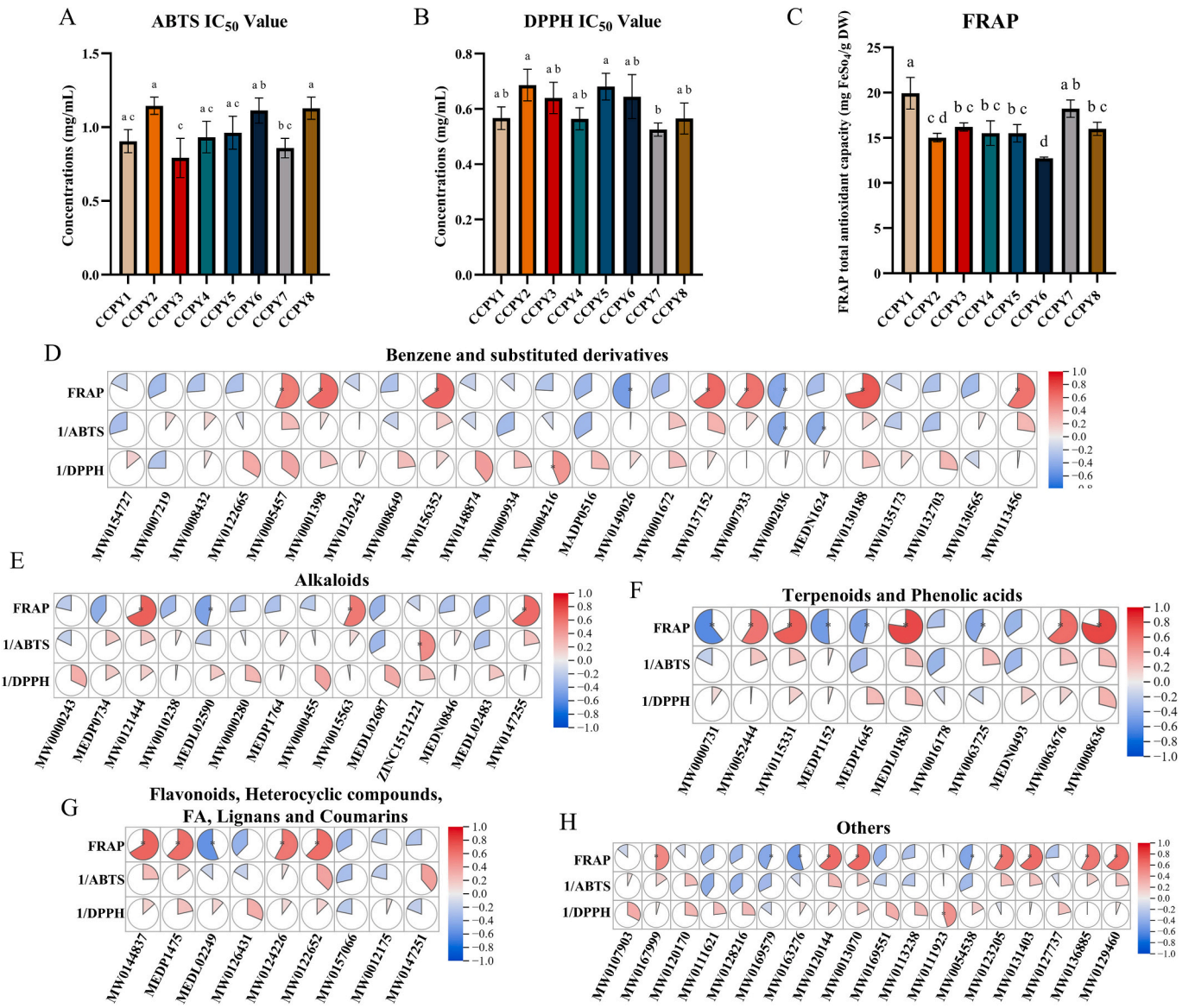


Fig. 7. Antioxidant activity and correlation analysis. (A) IC$_{50}$ value for ABTS radical scavenging activity. (B) IC$_{50}$ value for DPPH radical scavenging activity. (C) Total antioxidant capacity determined by the FRAP assay. (D—H) Correlation analysis between various metabolite classes and antioxidant activities: (D) Benzene and substituted derivatives, (E) Alkaloids, (F) Terpenoids and Phenolic acids, (G) Flavonoids, Heterocyclic compounds, FA, Lignans and Coumarins, and (H) Others. Different letters in the graphs indicate significant differences ($P < 0.05$). Compound numbers correspond to identifiers in the MetWare database, with full details provided in Table S4. * indicates a statistically significant correlation. All tests were repeated three times (n = 3).

Although certain constituents may play a dominant role, the contribution of other compounds should not be overlooked.

Correlation analysis between the antioxidant activities and the 76 shared DAMs was performed. (Note: The reciprocal of the IC$_{50}$ value was used for ABTS and DPPH assays.) As shown in Fig. 7D-H, most metabolites exhibited a positive correlation, with only a minority showing a negative correlation. Furthermore, a greater number of compounds demonstrated significant correlations with FRAP activity. This is likely because the ABTS and DPPH radical scavenging activities did not show significant variation among some of the CCP samples aged 1–8 years. It should be noted that key CRP constituents such as nobiletin, tangeretin, and hesperidin were excluded from the correlation analysis. This decision was made because the focus of this investigation was to identify the DAMs responsible for the variations in antioxidant activity among CCP samples, although these compounds are recognized as particularly important in CRP. This observation further supports the notion that the health-promoting effects of plant-based foods are not attributable to one single or a single class of compounds, but rather to the complex interplay of multiple constituents working in concert.

4. Conclusion

Here, we investigated the changes in the main physicochemical properties, volatile and non-volatile components, as well as antioxidant activity of CCP during 1–8 years of aging. The results showed that the flavonoid content in CCP initially increased and then decreased during the aging process, reaching a relatively high level in the fourth year. With prolonged aging time, the surface color of CCP gradually darkened, and the response intensity of the electronic nose sensors to CCP decreased. The FTIR results indicated that the aging process did not alter the types of compounds present in CCP, nor did it lead to the formation of any new functional groups. A total of 55 VOCs were identified in all samples by GC-IMS, most of which decreased in content during aging. As for non-volatile components, 1343 secondary metabolites were

identified using non-targeted metabolomics. Although the composition of metabolites did not change significantly during aging, their contents exhibited dynamic fluctuations. Through OPLS-DA analysis, 76 shared DAMs were screened across all comparison groups, which can serve as key biomarkers for determining the aging duration of CCP. In terms of bioactivity, the antioxidant activity first increased and then decreased. In conclusion, our study demonstrates that significant changes occur in the color, metabolite contents, and biological activities of CCP during the aging process. The most prominent changes take place within the first four years, indicating that the fourth year serves as a critical turning point for CCP quality evolution. Meanwhile, this study also reveals that within a limited timeframe, the quality improvement of CCP induced by aging may not follow a linear trend but rather exhibits a fluctuating pattern. This study provides a scientific basis for determining the aging period of CCP, facilitating the establishment of a tiered storage system. This system not only helps reduce long-term storage costs but also improves the utilization efficiency of raw material resources. Furthermore, the research transforms the traditional experience-based approach for assessing the aging year into an objective evaluation method based on chemical composition content, offering valuable references for market supervision and management. However, our research still lacks in vitro or in vivo validation of these activities, which will be the focus of our subsequent work. In addition, microorganisms also play a crucial role in the aging process. Our current study has not yet elucidated the effects of microorganisms on CCP aging, and subsequent research will be conducted from a microbial perspective to further unravel the aging mechanism of CCP. The current study can provide valuable insights for the aging process of CCP and offer a theoretical foundation for the development and production of CCP-based health foods.

CRediT authorship contribution statement

Jialiang Zou: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Huangsheng Zhang:** Writing – review & editing, Validation, Data curation. **Liang Zhao:** Writing – review & editing, Methodology, Data curation. **Huanhuan Xu:** Writing – review & editing, Methodology, Data curation. **Lin Chen:** Writing – review & editing, Methodology, Data curation. **Hongping Chen:** Writing – review & editing, Methodology, Data curation. **Fu Wang:** Writing – review & editing, Methodology, Data curation. **Yuan Hu:** Writing – review & editing, Validation, Funding acquisition, Formal analysis. **Youping Liu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2026.149212.

Data availability

Data will be made available on request.

References

Bender, E. (2022). Finding medical value in mescaline. *Nature, 609*(7929), S90–s91. https://doi.org/10.1038/d41586-022-02873-8

Bian, X., Xie, X., Cai, J., Zhao, Y., Miao, W., Chen, X., … Wu, J. L. (2022). Dynamic changes of phenolic acids and antioxidant activity of citri reticulatae pericarpium during aging processes. *Food Chemistry, 373*(Pt A), Article 131399. https://doi.org/10.1016/j.foodchem.2021.131399

Cai, Y., Ma, W., Hou, Y., Nisar, M. F., Long, Y., Chen, B., … Wan, C. (2025). Citri reticulatae pericarpium modulates gut microbiota: Impacts on human health. *Food Research International, 217*, Article 116745. https://doi.org/10.1016/j.foodres.2025.116745

Chen, X., Chen, H., Xiao, J., Liu, J., Tang, N., & Zhou, A. (2020). Variations of volatile flavour compounds in finger citron (citrus medica l. var. sarcodactylis) pickling process revealed by e-nose, HS-SPME-GC-MS and HS-GC-IMS. *Food Research International, 138*(Pt A), Article 109717. https://doi.org/10.1016/j.foodres.2020.109717

Chen, Y., Wen, B., Zhang, H., Chen, X., Bei, Q., Wang, Y., … Hu, Z. (2025). Comparative evaluation of flavor-related chemical compounds of 'feizixiao' litchi from various regions using e-tongue, e-nose, and HS-SPME-GC-MS. *Food Chemistry, 495*(Pt 3), Article 146560. https://doi.org/10.1016/j.foodchem.2025.146560

Chinese Pharmacopoeia Commission. (2020). *Pharmacopoeia of the People's Republic of China* (Volume I, pp. 199–200). China Medical Science and Technology Press.

Courdavault, V., & Papon, N. (2024). Plant drugs: Transcending the mescaline biosynthesis. *Current Biology, 34*(16), R791–r793. https://doi.org/10.1016/j.cub.2024.06.061

Dai, Q., Luo, W., Ye, Q., Chen., L, Hu, Y., Qin, K., Gao, P., & Luo, X. (2024). Research progress on changes in chemical composition and pharmacological effects of tangerine peel during microbial fermentation. Asia-Pacific Traditional Medicine. 20 (10), 250–255. https://link.cnki.net/urlid/42.1727.R.20240708.1009.008.

Deng, F., Chen, J., Fu, L., Zhong, J., Qiaoi, W., Luo, J., Li, J., & Li, N. (2024). Real-time citrus variety detection in orchards based on complex scenarios of improved YOLOv7. *Frontiers in Plant Science, 15*, Article 1381694. https://doi.org/10.3389/fpls.2024.1381694

Fu, M., Wang, Y., Yu, Y., Wen, J., Cheong, M. S., Cheang, W. S., & Wu, J. (2022). Changes of volatile substance composition during processing of nine-processed tangerine peel (jiuzhi chenpi) determined by gas chromatography-ion mobility spectrometry. *Frontiers in Nutrition, 9*, Article 963655. https://doi.org/10.3389/fnut.2022.963655

Fu, Y., Wang, C., Gao, Z., Liao, Y., Li, Z., Xiao, W., … Shan, Y. (2025). Decoding the role of bacillus: Unraveling the transformation of bioactive compounds during the aging of pericarpium citri reticulatae. *Food Research International, 214*, Article 116652. https://doi.org/10.1016/j.foodres.2025.116652

Gandhi, G. R., Vasconcelos, A. B. S., Wu, D. T., Li, H. B., Antony, P. J., Li, H., … Gan, R. Y. (2020). Citrus flavonoids as promising phytochemicals targeting diabetes and related complications: A systematic review of in vitro and in vivo studies. *Nutrients, 12*(10), 2907. https://doi.org/10.3390/nu12102907

Gao, X., Liu, Y., Chen, Y., Zhou, J., & Qu, D. (2014). Application of enzyme technology in flavonoid ingredients of chinese materia medica. *Chinese Traditional and Herbal Drugs, 45*(16), 2412–2417. https://doi.org/10.7501/j.issn.0253-2670.2014.16.025

Gerhardt, N., Schwolow, S., Rohn, S., Pérez-Cacho, P. R., Galán-Soldevilla, H., Arce, L., & Weller, P. (2019). Quality assessment of olive oils based on temperature-ramped HS-GC-IMS and sensory evaluation: Comparison of different processing approaches by LDA, kNN, and SVM. *Food Chemistry, 278*, 720–728. https://doi.org/10.1016/j.foodchem.2018.11.095

He, T., Gou, H., Wang, B., Wang, F., Chen, H., & Liu, Y. (2025). Aging mechanism of pericarpium citri reticulatae 'chachiensis' during in situ storage based on widely TargetedMetabolomics and high-throughput sequencing. *Food Science, 46*(16), 33–43. https://link.cnki.net/urlid/11.2206.TS.20250331.1250.058.

Hern´andez-Mesa, M., Escourrou, A., Monteau, F., Le Bizec, B., & Dervilly-Pinel, G. (2017). Current applications and perspectives of ion mobility spectrometry to answer chemical food safety issues. *TrAC - Trends in Analytical Chemistry, 94*, 39–53. https://doi.org/10.1016/j.trac.2017.07.006

Hsu, T. Y., Yang, K. M., Chiang, Y. C., Lin, L. Y., & Chiang, P. Y. (2024). The browning properties, antioxidant activity, and α-glucosidase inhibitory improvement of aged oranges (citrus sinensis). *Foods, 13*(7). https://doi.org/10.3390/foods13071093

Hu, Y., Hu, D., Yin, L., Deng, Z., Cheng, Y., Li, H., Wang, F., & Liu, Y. (2025). Dynamic profiling of bioactive compounds, flavor metabolites, and quality-related microorganisms during the freshening-drying-aging process of citri reticulatae pericarpium: Implications for quality formation mechanisms. *Food Chemistry X, 30*, Article 102906. https://doi.org/10.1016/j.fochx.2025.102906

Hu, Y., Wu, B., Yi, D., Chen, H., Yang, F., Wang, F., & Liu, Y. (2022). Integration of high throughput sequencing and widely targeted metabolomics reveals the aging mechanism of pericarpium citri reticulatae 'chachiensis'. *Natural Product Research and Development, 04*, 553–562. https://doi.org/10.16333/j.1001-6880.2022.4.002

Jia, X., Zhang, S., Lin, X., Nie, X., Kang, X., Pu, H., … Zhao, Z. (2025). Physicochemical properties, aroma compounds, non-volatile metabolites and their potential correlation in wickerhamomyces anomalus-fermented Sichuan-style sausage. *Food Chemistry X, 29*, Article 102642. https://doi.org/10.1016/j.fochx.2025.102642

Jiang, K., Xu, K., Wang, J., Meng, F., & Wang, B. (2023). Based on HS-SPME-GC-MS combined with GC-O-MS to analyze the changes of aroma compounds in the aging

process of *citri reticulatae* pericarpium. Food Bioscience, 54, Article 102798. https://doi.org/https://doi.org/10.1016/j.fbio.2023.102798.
Lan, Z., Zhang, X., Cai, X., Pan, W., Li, W., Sun, Y., … Meng, J. (2025a). Metabolite profiling and 'dryness'-like effect analysis of 'chenpi' plant tea (citri reticulatae pericarpium peel) with and without steaming peocessing. *Food Research International, 206*, Article 116089. https://doi.org/10.1016/j.foodres.2025.116089
Lan, Z., Zhang, X., Cai, X., Pan, W., Li, W., Sun, Y., … Meng, J. (2025b). Metabolite profiling and "dryness"-like effect analysis of "chenpi" plant tea (citri reticulatae pericarpium peel) with and without steaming peocessing. *Food Research International, 206*, Article 116089. https://doi.org/10.1016/j.foodres.2025.116089
Li, G., & Facchini, P. J. (2024). New frontiers in the biosynthesis of psychoactive specialized metabolites. *Current Opinion in Plant Biology, 82*, Article 102626. https://doi.org/10.1016/j.pbi.2024.102626
Li, S., Mao, X., Guo, L., & Zhou, Z. (2023). Comparative analysis of the impact of three drying methods on the properties of *citrus reticulata* Blanco cv. Dahongpao Powder and Solid Drinks. *Foods, 12*(13). https://doi.org/10.3390/foods12132514
Li, S. Z., Zeng, S. L., Wu, Y., Zheng, G. D., Chu, C., Yin, Q., … Liu, E. H. (2019). Cultivar differentiation of citri reticulatae pericarpium by a combination of hierarchical three-step filtering metabolomics analysis, DNA barcoding and electronic nose. *Analytica Chimica Acta, 1056*, 62–69. https://doi.org/10.1016/j.aca.2019.01.004
Li, Y., Zhao, W., Qian, M., Wen, Z., Bai, W., Zeng, X., … Dong, H. (2024). Recent advances in the authentication (geographical origins, varieties and aging time) of tangerine peel (citri reticulatae pericarpium): A review. *Food Chemistry, 442*, Article 138531. https://doi.org/10.1016/j.foodchem.2024.138531
Liang, S., Wen, Z., Tang, T., Liu, Y., Dang, F., Xie, T., & Wu, H. (2022). Study on flavonoid and bioactivity features of the pericarp of citri reticulatae 'chachi' during storage. *Arabian Journal of Chemistry, 15*(3), Article 103653. https://doi.org/10.1016/j.arabjc.2021.103653
Liu, H., Fu, M., Wu, J., Yu, Y., Si, W., Xu, Y., & Yang, J. (2025). Flavor characterization of aged citri reticulatae pericarpium from core regions: An integrative approach utilizing GC-IMS, GC-MS, e-nose, e-tongue, and chemometrics. *Food Chemistry, 490*, Article 144995. https://doi.org/10.1016/j.foodchem.2025.144995
Liu, N., Li, X., Zhao, P., Zhang, X., Qiao, O., Huang, L., Guo, L., & Gao, W. (2021). A review of chemical constituents and health-promoting effects of citrus peels. *Food Chemistry, 365*, Article 130585. https://doi.org/10.1016/j.foodchem.2021.130585
Liu, N., Yang, W., Li, X., Zhao, P., Liu, Y., Guo, L., Huang, L., & Gao, W. (2022). Comparison of characterization and antioxidant activity of different citrus peel pectins. *Food Chemistry, 386*, Article 132683. https://doi.org/10.1016/j.foodchem.2022.132683
Liu, Y., Wen, H., Kong, J., Hu, Z., Hu, Y., Zeng, J., … Xu, J. (2024). Flavor characterization of citri reticulatae pericarpium (citrus reticulata 'chachiensis') with different aging years via sensory and metabolomic approaches. *Food Chemistry, 443*, Article 138616. https://doi.org/10.1016/j.foodchem.2024.138616
Luo, X., Kang, S., Lei, L., Gao, P., Xu, L., Yang, X., … Ma, S. (2023). Application of citri reticulatae pericarpium from Sichuan ("dahongpao"). *Chinese Pharmaceutical Affairs, 02*, 178–186. https://doi.org/10.16153/j.1002-7777.2023.02.008
Ma, D., Mi, L., Xu, S., Lü, S., & Zhong, B. (2022). Comparative analysis of the aroma components of the peel of six citrus varieties in southern Ganzhou. Molecular Plant Breeding, 1-19. Doi: //link.cnki.net/urlid/46.1068.s.20220309.1619.008.
Mamy, D., Boateng, I. D., & Chen, X. (2025). Metabolomic changes in citrus reticulata peel after conventional and ultrasound-assisted solid-state fermentation with aspergillus Niger: A focus on flavonoid metabolism. *Food Chemistry, 467*, Article 142224. https://doi.org/10.1016/j.foodchem.2024.142224
Mamy, D., Huang, Y., Akpabli-Tsigbe, N. D. K., Battino, M., & Chen, X. (2022). Valorization of citrus reticulata peels for flavonoids and antioxidant enhancement by solid-state fermentation using aspergillus Niger CGMCC 3.6189. *Molecules, 27*(24). https://doi.org/10.3390/molecules27248949
Meng, L., Nie, Y., Wu, S., Li, W., Zhang, B., Tao, G., … Xu, S. (2026). Effects of variety and cultivation on volatile organic compounds in dictyophora rubrovolvata: Insights from integrated GC-IMS, GC-MS and e-nose and multivariate data analysis. *Food Chemistry: X, 33*, Article 103516. https://doi.org/10.1016/j.fochx.2026.103516
Milke, L., Aschenbrenner, J., Marienhagen, J., & Kallscheuer, N. (2018). Production of plant-derived polyphenols in microorganisms: Current state and perspectives. *Applied Microbiology and Biotechnology, 102*(4), 1575–1585. https://doi.org/10.1007/s00253-018-8747-5
Payne, J. T., Valentic, T. R., & Smolke, C. D. (2021). Complete biosynthesis of the bisbenzylisoquinoline alkaloids guattegaumerine and berbamunine in yeast. *Proceedings of the National Academy of Sciences of the United States of America, 118*(51). https://doi.org/10.1073/pnas.2112520118
Peng, M., Gao, Z., Liao, Y., Guo, J., & Shan, Y. (2022). Development of Citrus-based functional jelly and an investigation of its anti-obesity and antioxidant properties. *Antioxidants (Basel, Switzerland), 11*(12), 2418. https://doi.org/10.3390/antiox11122418
Qin, Y., Zhao, Q., Zhou, D., Shi, Y., Shou, H., Li, M., Zhang, W., & Jiang, C. (2024). Application of flash GC e-nose and FT-NIR combined with deep learning algorithm in preventing age fraud and quality evaluation of pericarpium citri reticulatae. *Food Chemistry X, 21*, Article 101220. https://doi.org/10.1016/j.fochx.2024.101220
Shi, J., Peng, L., Chen, W., Qiao, W., Wang, K., Xu, Y., & Cheng, J. (2024). Evaluation of chemical components and quality in xinhui chenpi (citrus reticulata "chachi") with two different storage times by GC-MS and UPLC. *Food Science & Nutrition, 12*(7), 5036–5051. https://doi.org/10.1002/fsn3.4154
Shorbagi, M., Fayek, N. M., Shao, P., & Farag, M. A. (2022). Citrus reticulata Blanco (the common mandarin) fruit: An updated review of its bioactive, extraction types, food quality, therapeutic merits, and bio-waste valorization practices to maximize its economic value. *Food Bioscience, 47*, Article 101699. https://doi.org/10.1016/j.fbio.2022.101699
Singh, A., & Negi, P. S. (2025). Multivariate analysis of the efficiency of bio-actives extraction from laboratory-generated by-products of *citrus reticulata* Blanco. *Journal of Food Science, 90*(6), Article e70304. https://doi.org/10.1111/1750-3841.70304
Song, Y., Chen, H., Tan, S., Hu, Y., & Yang, W. (2017). A comparative study on content of hesperidin, nobiletin, tangeretin and synephrine in pericarpium citri reticulatae from different varieties. Journal of Li-Shizhen Traditional Chinese Medicine, 09, 2061–2064. Doi: CNKI:SUN:SZGY.0.2017-09-005.
Wang, F., Hu, Y., Chen, H., Chen, L., & Liu, Y. (2023). Exploring the roles of microorganisms and metabolites in the 30-year aging process of the dried pericarps of citrus reticulata 'chachi' based on high-throughput sequencing and comparative metabolomics. *Food Research International, 172*, Article 113117. https://doi.org/10.1016/j.foodres.2023.113117
Wang, F., Zhang, X., Lu, J., Mei, G., Chen, H., Chen, L., & Liu, Y. (2015). Study of reason of "the older,the better"of Pericarpium Citri Reticulatae(PCR) which contributed increase of flavonoids. China Journal of Chinese Materia Medica, 24, 4890–4896. Doi: CNKI:SUN:ZGZY.0.2015-24-028.
Wang, H., Wang, P., Wang, F., Chen, H., Chen, L., Hu, Y., & Liu, Y. (2024). Integrated HS-GC-IMS and UPLC-Q-orbitrap HRMS-based metabolomics revealed the characteristics and differential volatile and nonvolatile metabolites of different citrus peels. *Current Research in Food Science, 8*, Article 100755. https://doi.org/10.1016/j.crfs.2024.100755
Wang, J., Chen, H., Liu, Y., Liu, R., & Fan, D. (2013). Chemical variability of essential oils in xinhui citri reticulatae pericarpium stored in different years. Journal of Li-Shizhen Traditional Chinese Medicine, 12, 2831–2834. Doi: CNKI:SUN:SZGY.0.2013-12-003.
Wang, P., Wang, H., Zou, J., Chen, L. i., Chen, H., Hu, Y., Wang, F., & Liu, Y. (2023). Electronic nose and head space GC-IMS provide insights into the dynamic changes and regularity of volatile compounds in zangju (*citrus reticulata* cv. Manau Gan) peel at different maturation stages. Molecules, 28(14). Doi: https://doi.org/10.3390/molecules28145326.
Wei, Z., Chen, H., Yang, L., Liu, R., Peng, Y., & Liu, Y. (2013). Study on dynamic change law of synephrine and total flavone in citrus reticulate chachi from different storage periods. *Liaoning Journal of Traditional Chinese Medicine, 05*, 982–985. https://doi.org/10.13192/j.ljtcm.2013.05.156.weizh.069
Wu, T., Cheng, D., He, M., Pan, S., Yao, X., & Xu, X. (2014). Antifungal action and inhibitory mechanism of polymethoxylated flavones from citrus reticulata Blanco peel against aspergillus Niger. *Food Control, 35*(1), 354–359. https://doi.org/10.1016/j.foodcont.2013.07.027
Xu, G., Xu, L., & Wang, Z. (2001). *Studies on the identity and quality of commonly used Chinese medicinal materials: Southern collaborative group (volume 4)* (p. 467). Fujian Science and Technology Publishing House.
Yang, F. (2022). Study the development of the microorganisms and flavonoids changes of citrus reticulata 'chachi' during aging. *Chengdu University of Traditional Chinese Medicine.* https://link.cnki.net/doi/10.26988/d.cnki.gcdzu.2021.000736.
Yang, F., He, L., Shen, M., Wang, F., Chen, H., & Liu, Y. (2022). A correlation between pericarpium citri reticulatae volatile components and the change of the coexisting microbial population structure caused by environmental factors during aging. *Frontiers in Microbiology, 13*, Article 930845. https://doi.org/10.3389/fmicb.2022.930845
Yang, F., He, Liying, Yang, D., Shen, M., Chen, H.-p., & Liu, Y. (2021). Analysis and identification of flavonoids in citri reticulatae pericarpium with different aging time by UPLC-Q-orbitrap HRMS. Chinese Journal of Experimental Traditional Medical Formulae, 12, 125–132. Doi: 10.13422/j.cnki.syfjx.20210249.
Yao, W., Ma, S., Wu, H., Liu, D., Liu, J., & Zhang, M. (2024). Flavor profile analysis of grilled lamb seasoned with classic salt, chili pepper, and cumin (cuminum cyminum) through HS-SPME-GC-MS, HS-GC-IMS, e-nose techniques, and sensory evaluation on sonit sheep. *Food Chemistry, 454*, Article 139514. https://doi.org/10.1016/j.foodchem.2024.139514
Yin, J., Wu, M., Lin, R., Li, X., Ding, H., Han, L., … Li, Z. (2021). Application and development trends of gas chromatography–ion mobility spectrometry for traditional Chinese medicine, clinical, food and environmental analysis. *Microchemical Journal, 168*. https://doi.org/10.1016/j. microc.2021.106527
Yu, X., Sun, S., Guo, Y., Liu, Y., Yang, D., Li, G., & Lü, S. (2018). Citri reticulatae pericarpium (chenpi): Botany, ethnopharmacology, phytochemistry, and pharmacology of a frequently used traditional chinese medicine. *Journal of Ethnopharmacology, 220*, 265–282. https://doi.org/10.1016/j.jep.2018.03.031
Zeng, H., Zeng, L., Chen, J., Chen, X., Peng, R., Huang, Z., Zeng, X., & Zeng, Z. (2023). Research progress on chemical constituents difference of *citri reticulatae pericarpium* with different storage years. Asia-Pacific Traditional Medicine, 19(07), 221–225. Doi: CNKI:SUN:YTCT.0.2023-07-045.
Zhang (2025). Research progress on the aging mechanism and biological activity of tangerine peel. Food Science 1–15. https://link.cnki.net/urlid/11.2206.TS.20251023.1617.020.
Zhang, H., Xie, Y., Liu, C., Chen, S., Hu, S., Xie, Z., Deng, X., & Xu, J. (2017). Comprehensive comparative analysis of volatile compounds in citrus fruits of different species. *Food Chemistry, 230*, 316–326. https://doi.org/10.1016/j.foodchem.2017.03.040
Zhang, J. (2018). Study on the evaluation of quality between pericarpium citri reeticulatae 'chachi' and other 5 species citri reticulatae. Guangzhou University of Chinese Medicine. https://kns.cnki.net/kcms2/article/abstract?v=Mz9udXFtQxk7u-9dIf6x_XuC1wCgUvnq-UZrJIG9HPiZSgDYyVpQzjCoq_4_L4eiWnWzwETXvjZNupKIy38l3US4HBTxlidCHLJMWYp8w4x0167DQwMg72d7qvmEEZCyYqYG6jRtar-qnlMdJioRqMElm1O7AIiN1zn7YbFBrg-Ajq-EPpa3WUR_yb9kfAZP&uniplatform=NZKPT&language=CHS.
Zhang, J., Wu, X., Qiu, J., Zhang, L., Zhang, Y., Qiu, X., … Xu, W. (2020). Comprehensive comparison on the chemical profile of guang chen pi at different ripeness stages

using untargeted and pseudotargeted metabolomics. *Journal of Agricultural and Food Chemistry, 68*(31), 8483–8495. https://doi.org/10.1021/acs.jafc.0c02904

Zhang, J.-B., Li, M.-X., Zhang, Y.-F., Qin, Y.-W., Li, Y., Su, L.-L., … Lu, T.-L. (2023). E-eye, flash GC e-nose and HS-GC-MS combined with chemometrics to identify the adulterants and geographical origins of ziziphi spinosae semen. *Food Chemistry, 424*, Article 136270. https://doi.org/10.1016/j.foodchem.2023.136270

Zhang, Y., Yan, Y., Shu, H., Dan, Q., Pan, L., & Tu, K. (2025). Analysis of aroma and key differential volatile compounds during ripening stages of three strawberry cultivars using HS-SPME-GC-MS, GC-IMS, and e-nose. *Food Chemistry, 494*, Article 146199. https://doi.org/10.1016/j.foodchem.2025.146199

Zhao, H., Chen, H., Wu, G., Xu, J., Zhu, W., Chen, J., … Guo, S. (2025). Integrative metabolomics - GC-IMS approach to assess metabolic and flavour substance shifts during fermentation of Yangjiang douchi. *Food Chemistry, 466*, Article 142199. https://doi.org/10.1016/j.foodchem.2024.142199

Zheng, G., Chao, Y., Liu, M., Yang, Y., Zhang, D., Wang, K., … Wei, M. (2021). Evaluation of dynamic changes in the bioactive components in citri reticulatae pericarpium (citrus reticulata "chachi") under different harvesting and drying conditions. *Journal of the Science of Food and Agriculture, 101*(8), 3280–3289. https://doi.org/10.1002/jsfa.10957

Zou, J., Wang, P., Xu, H., Gan, X., Zhang, H., Chen, L., … Liu, Y. (2024). Metabolic profile and bioactivity of the peel of zhoupigan (*citrus reticulata* cv. Manau Gan), a special citrus variety in China, based on GC-MS, UPLC-ESI-MS/MS analysis, and in vitro assay. *Food chemistry x, 23*, Article 101719. https://doi.org/10.1016/j.fochx.2024.101719